

Credential Rationing and Competition

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Abstract

This paper develops a theory of *credential rationing* when a privately informed sender relies on an intermediary to persuade receivers. The intermediary sells tests to the sender, who possesses imperfect private information about his quality and seeks to persuade the receivers to match with him. Screening the sender's private information distorts the allocation of tests and matches. To economize on information rents, the intermediary restricts the supply of high credentials by steering allocation toward less selective receivers. Furthermore, this rationing can shift matches toward the bottom receiver and reduce the top receiver's share relative to the no-screening benchmark.

KEYWORDS: Monopoly Certification, Information Acquisition, Mechanism Design, Communication Game, Adverse Selection

JEL CLASSIFICATIONS C70, D45, D82, J44

1 Introduction

Markets are subject to information asymmetries as senders generally know more about their quality than potential receivers. This asymmetry can lead to market failure if the sender cannot credibly signal his private information and if the expected quality of the sender is low, as in [Akerlof \(1970\)](#). An institutional response to market failure is third-party certification. The certifier allows the informed sender to signal information to uninformed receivers (decision makers). The certifier sells tests to the sender, whose outcome may depend on the sender's underlying quality. The test outcome informs the receivers about the sender's quality, alleviating some of the asymmetry.

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Labor markets provide one concrete application: professional examinations and other certification tests convey information about applicants' skills. Employers often screen applicants through an established minimum *hiring standard*.¹ These can take the form of pass/fail requirements or requirements on test scores. When wages are standardized, employers compete in their selectivity to attract applicants. More sought-after employers can screen applicants by setting higher hiring standards.

The institutions responsible for administering these tests face an incentive problem in allocating tests to privately informed senders. Rent-extraction by the certifier can lead to distortions in test allocation, which feeds back into the receivers' decisions. To parse the effect of the certifier's rent-extracting motive on the test allocation and on the subsequent receiver-sender interaction, I abstract away from institutional details and consider a revenue-maximizing certifier that can flexibly design and price information through menu pricing of tests.

In that respect, this paper connects two elements that are typically studied in isolation: the market structure for allocating tests to the sender and the downstream receiver-sender interaction. This provides new theoretical insights about the connection between allocative efficiency in the certifier-sender market and the nature of competition between receivers. In particular, second-degree price discrimination by the certifier can shift allocations toward less selective receivers.² This can increase the bottom receiver's share of matches and reduce the top receiver's share relative to the benchmark.

Rent-extraction by the certifier restricts the supply of high-credential senders, a distortion I call *credential rationing*, and thereby shapes the allocation of matches across receivers.

Certification and Standards: The certifier can be viewed as a two-sided supplier. The certifier sells tests to a privately informed sender. The certifier "indirectly supplies" certified senders to receivers.³ The certifier can utilize menus of tests to (second-degree) price discriminate the privately informed sender. A receiver makes the acceptance decision based on the sender's testing outcome and the menu of testing options available to the sender. Tests contain hard information, as the outcomes depend on the sender's underlying quality. Tests also contain soft information owing to the sender's self-selection into different testing options made available by the certifier.

Receivers can set high standards to drive out lower quality senders when testing is sufficiently informative and costly. Although a convenient screening tool, an excessively high acceptance

¹A hiring standard represents the minimum expected ability of an applicant that an employer is willing to hire.

²Lizzeri (1999) and Kartik et al. (2021) show that monopolistic certification, without screening, has an *informativeness-reducing* effect. I expand on this insight by showing that second-degree price discrimination by the certifier further reduces informativeness. Here, allocating to a less selective receiver allows greater pooling of low-quality senders and reduces information rents.

³An alternate setting could allow the certifier to charge a price to the receivers for revealing a sender's test outcome. In this case, the certifier can be viewed as a direct supplier to the receiver. I comment on this in section 7.2.

standard can be counterproductive when receivers compete for a limited pool of senders. A competing receiver might be able to poach potential high quality senders by undercutting a receiver with stricter standards. These considerations highlight the strategic role of acceptance standards in addition to the aforementioned screening role.

Another strategic aspect of acceptance standards is their ability to influence the tests designed by the certifier. As the sender's willingness to pay for a test depends on the receivers' acceptance decisions, the acceptance standards also affect the demand for certification tests.⁴ This way, receivers' acceptance standards and the test allocation are jointly determined in an equilibrium. Due to this feedback, inefficiencies in test allocation can lead to unexpected consequences for receiver competition.

Modeling Preliminaries: I consider a monopolistic certifier (test designer), a sender who is partially and privately informed about his quality, and two receivers that demand certified senders and are differentiated by their reputation.^{5,6}

The model abstracts away from price-setting to focus on certification's impact on receiver competition. In the labor-market application, this corresponds to taking wages as given while allowing applicants to prefer some employers over others because of nonwage aspects such as job mobility or workplace environment.⁷

I model imperfect receiver competition through two vertically differentiated receivers. I consider two receivers that differ in their utility to the sender from a match. I call them the top receiver and the bottom receiver. The utility from joining the top receiver is greater than that from the bottom receiver, and this is the same across all senders. Receivers rely on test outcomes to assess the quality of a potential sender. Both receivers have the same value for a sender of a given quality; receivers have a higher utility for higher quality senders.

To focus on receiver competition, I assume receivers have greater bargaining power than the certifier. In particular, receivers can commit to an acceptance standard rather than choosing their standards in response to the mechanism designed by the certifier. Receivers affect the demand for certified senders by committing to acceptance standards.

The sender is partially privately informed about his quality. I refer to the sender's belief about his quality as his type. As test outcomes depend on quality, the sender's value for any given test

⁴Information is valuable only through the decisions it induces.

⁵Many examples of professional certification, including the ones alluded to above, involve third-party certification that is not necessarily monopolistic. Yet the monopolistic certification is an economically relevant benchmark, and insights from the analysis are useful for other settings as well.

⁶The two-receiver structure isolates competition over acceptance standards.

⁷In section 7.1, I present a labor-market extension with wages. The main insight there is a negative result. If the certifier can flexibly design and price information (tests), then vertical differentiation of employers alone does not lead to meaningful wage competition.

varies with the belief about his quality. In particular, the sender's type determines his willingness to pay for a test.

The receivers and the certifier do not observe the sender's type (and quality), and have a common prior F . I assume that in the absence of the certifier, there is a market failure, as the sender is always left unmatched. In particular, I assume that the sender's prior expected quality is below the receivers' reservation cutoff for a match. This makes the certifier's ability to generate hard information (quality-contingent signaling) central in preventing market failure.

Due to uncertainty about the sender's type, the certifier faces a screening problem when selling tests to the sender. Tests are multi-dimensional instruments, each comprising a collection of outcome distributions indexed by the sender's underlying quality.⁸ The variation in a test's value across different types of senders depends on the variation in the test's quality contingent outcome distributions. I restrict attention to binary quality to simplify the analysis of the screening problem faced by the certifier. The sender is either high quality or low quality. The sender's type is then captured by the probability he assigns to being high quality. Types are single-dimensional, sidestepping intricacies associated with the design of selling mechanisms in the presence of multi-dimensional type spaces.

Along with the screening problem above, the certifier faces an information design problem. Tests are valuable to a sender only if they persuade the receiver to accept the sender. Acceptance standards constrain the information a test must generate to persuade a receiver. This restricts the certifier's ability to pool high and low quality senders, thus constraining the surplus that the certifier can generate. The certifier needs to trade off surplus generation and information rents from screening friction, while conforming to acceptance standards set by the receivers.

The interaction of the screening and information design problems results in inefficiencies that are characteristic of certification markets. The certifier reduces the informativeness of tests by pooling low quality senders with high quality senders to economize on information rents. As the certification mechanism and acceptance standards are equilibrium objects, these inefficiencies spill over onto the receivers' decisions.

Results: In the model, screening frictions faced by the certifier lead to two distortions relative to a benchmark where the certifier observes the sender's type at the time of contracting. The benchmark captures the information design part of the certifier's problem (see section 4).

The first distortion is *exclusion*; there is an increase in the chance that the sender is left unmatched. When facing a privately informed sender, the certifier resorts to second-degree price discrimination. This involves offering a menu of different testing options at varying prices. The optimal selling mechanism for tests leads to exclusion whenever the expected quality of the senders

⁸In economics and statistics literature such tests are often referred to as Blackwell (or statistical) experiments.

is low enough. In the presence of screening frictions, the designer does not sell to some types, increasing exclusion.⁹ For exclusion to occur, it's crucial that the certifier needs to generate informative tests to induce a match. There is an excessive reduction in the supply of certified senders, relative to the benchmark, indicating that the inefficiency in test allocation amplifies certification's role as a barrier to entry for the sender.

The second distortion is *allocation toward less selective receivers*; the certifier distorts the test towards allocating the sender to less selective receivers with a greater probability. Less selective receivers set lower acceptance standards and are willing to accept senders with lower expected quality. Pooling a larger quantity of low quality senders helps reduce overall information rents, as it reduces the difference (heterogeneity) in test outcomes for high and low quality senders. The certifier benefits from a greater mass of low quality senders not only through changes in the sender's gross utility but also through reductions in information rents conceded to the sender. This *informativeness reducing* effect of second-degree price discrimination is highlighted by Theorem 1 in section 5.2. Theorem 1 shows that whenever the top receiver sets a higher standard than the bottom receiver, the certifier allocates the sender to the bottom receiver with positive probability (and sometimes allocates the sender only to the bottom receiver). This contrasts with the benchmark, where the certifier allocates the sender only to the top receiver against a subset of these standards.

Although tests that allocate the sender to the top receiver fetch higher prices, they also raise information rents disproportionately when the top receiver sets a higher standard than the bottom receiver. Higher standards result in a smaller quantity of low quality senders, leading to greater heterogeneity in willingness to pay across sender types and thus greater information rents. When the bottom receiver sets a low enough standard, the certifier might prefer allocating the sender to the bottom receiver, even when it reduces the sender's gross utility. This creates a channel through which the bottom receiver gains matches at the expense of the top receiver. I demonstrate this effect on receiver competition in Theorem 2. In the benchmark equilibria identified in Proposition 1, the sender matches with the bottom receiver with zero probability. When the certifier screens the sender, both receivers earn positive expected payoffs in every equilibrium. The top receiver's share of matches consequently falls relative to these benchmark allocations.

The results of this paper identify the role of monopolistic certification in shaping receiver competition by excessively reducing the supply of high-credential senders through reducing the informativeness of test allocation.

⁹This distortion is reminiscent of exclusion results from the literature on non-linear pricing, see [Armstrong \(1996\)](#) for example.

2 Literature

Theoretical and empirical literature on certification is vast; an early contribution by [Viscusi \(1978\)](#) points out the role of certification in preventing market breakdown. In a seminal paper, [Lizzeri \(1999\)](#) explores the role of a monopolistic certifier who can sell information to privately informed parties. But the designer in [Lizzeri \(1999\)](#) uses a take-it-or-leave-it offer instead of a menu of tests and prices. [Lizzeri \(1999\)](#) assumes that markets do not unravel when the certifier is absent; the prior expected quality of the sender is above the receivers' reservation. Thus, restricting the certifier to a single take-it-or-leave-it offer is without loss as the certifier offers an uninformative test sold to all types and extracts all the surplus from the senders. [Kartik et al. \(2021\)](#) generalizes the findings of [Lizzeri \(1999\)](#) to partially informed agents; they do not allow price discrimination by the certifier. A key insight in these papers is that a monopolistic certifier favors less informative tests. I extend the insight of these papers by showing that second-degree price discrimination by the certifier amplifies the economic force favoring less information.

In an influential paper [Leland \(1979\)](#) develops a theory of professional licensing in an environment where applicants invest in their ability. He shows that licensing, in the form of minimum ability standards, can prevent market failure. [Leland \(1979\)](#) does not consider certification or an information intermediary in his model; the choice and enforcement of standards is exogenous.

A well-known consequence of certification is its function as a barrier to entry ([Stigler \(1971\)](#)). Certification requirements create restrictions on the supply of senders and potentially drive up competition between receivers. I argue that inefficiency in test allocation, from certifier screening the sender, not only further restricts the supply of senders but also reduces the overall informativeness of the tests.

[Dranove and Jin \(2010\)](#) surveys, among other aspects of certification, the economic debate about certification's role in *quality assurance* versus its role as a *barrier to entry*. [Naidu and Posner \(2022\)](#) surveys the challenges present in regulating employer competition. [Azar and Marinescu \(2024\)](#) surveys recent developments in the theory of employer competition. They focus on three modeling approaches for employer competition – oligopoly, job differentiation, and search frictions. The labor-market application of this paper addresses a different question about certification's quality-assurance role. How do distortions from screening affect competition among receivers when the certifier flexibly designs and prices information?

Following the works of [Rayo and Segal \(2010\)](#) and [Kamenica and Gentzkow \(2011\)](#), there has been an explosion of interest in studying information provision involving general information structures. This has led to new insights about markets and methodological advances [Roesler and Szentes \(2017\)](#), [Bergemann et al. \(2018\)](#), [Kleiner et al. \(2021\)](#), [Dworczak and Kolotilin \(2024\)](#). In a recent survey [Bergemann and Ottaviani \(2021\)](#) describes various market mechanisms for information provision. Using techniques from the information design literature, there have been recent

developments in the study of certification, where the certifier produces hard information. [Ali et al. \(2020\)](#) has considered robust provision of hard information when the agent does not hold private information. [Asseyer and Weksler \(2024\)](#) also considers the provision of hard information to uninformed agent; they focus on a common value environment.

Combining mechanism design and information design problems has led to many interesting insights in various applications. For example, [Calzolari and Pavan \(2006\)](#), [Dworczak \(2020\)](#) have studied sequential agency problems in which upstream designers reveal strategic information to downstream principals. [Bergemann and Pesendorfer \(2007\)](#), [Bergemann et al. \(2022\)](#) have considered joint design of product allocation and information about the product. [Frankel \(2021\)](#) considers a delegation model in a labor market environment where employers get hard and soft information about applicants. The important difference is that the employer can observe the applicant's hard information, and the soft information is provided by a third agent (manager). [Corrao \(2023\)](#) has studied screening by a monopolistic certifier, but restricts attention to soft information.

There has been some recent interest and progress in studying certification intermediaries that generate new information and interact with partially informed buyers [Weksler and Zik \(2025\)](#), [Celik and Strausz \(2025\)](#), [Mäkimattila et al. \(2025\)](#), [Chopra and Ely \(2025\)](#). In [Weksler and Zik \(2025\)](#), the agents seeking certification are privately and partially informed, but their choice of tests is publicly observable. [Celik and Strausz \(2025\)](#) focuses on the role of soft and hard information (they call it screening and acquisition) in certification mechanisms for a buyer seller framework. [Mäkimattila et al. \(2025\)](#) focuses on monopolistic certification, similar to my single receiver benchmark, and contrasts it with the setting in which the sender's test choice is observable. In a related single receiver model [Ely \(2025\)](#) studies optimal test allocation when the sender can credibly reveal his private type to the receiver (decision maker).

The closest paper to this one is [Chopra and Ely \(2025\)](#), which develops analytical tools for the mechanism and information design problems faced by the certifier in various contexts. Although both [Mäkimattila et al. \(2025\)](#) and [Chopra and Ely \(2025\)](#) describe the single receiver analogue of the certifier's problem (see section 6.1), neither considers the equilibrium interaction of the information receivers' commitment and certification mechanism.

3 Model

There are four players: a sender, a top receiver, a bottom receiver, and a certifier (test designer). The sender has quality $\theta \in \{h, l\}$, unknown to all. The sender has partial private information about his quality represented by his **type**, $\mu \in [0, 1]$. Only the sender knows his type, $\mu = \Pr(\theta = h \mid \text{private information})$. I use expected quality as shorthand for the probability of high quality. The receivers and the certifier are initially uninformed about the sender's quality θ and his type

μ , and have a common prior $F \in \Delta([0, 1])$ with full support and a continuous density f . I also assume that $f(1) > 0$.

Each receiver receives a payoff of $v_h > 0$ from matching with a high quality sender and a payoff of $v_l < 0$ from matching with a low quality sender. The value v_θ can be understood as the match value of a sender with quality θ . The expression $\nu v_h + (1 - \nu)v_l$ is the expected match value (value), to the receivers, of a sender with expected quality ν . The payoff from matching with a sender with expected quality ν is $\nu v_h + (1 - \nu)v_l$. Thus, the expected value of a sender to the receivers is increasing in the sender's type. Define the reservation cutoff by $q := -v_l / (v_h - v_l)$. The receivers are indifferent between accepting and rejecting a sender whose probability of high quality is q . The cutoff q is the receivers' *reservation* expected quality of the sender.

I will assume that $1 > q > \mathbb{E}_F[\mu]$. This makes the role of certifier more interesting, as without the certifier, there is no match. The assumption is to emphasize the certifier's role in information generation, in contrast to the gatekeeping role of certification. If $\mathbb{E}_F[\mu] < q$ then the certifier enables credible signaling which prevents market failure. The certifier is a gatekeeper when $\mathbb{E}_F[\mu] \geq q$. In this case, markets do not unravel without the certifier, as the sender's prior expected value to the receivers is positive.

Imperfect competition between receivers is modeled through vertical differentiation. The sender gets a utility of 1 from matching with the top receiver, a utility of $0 < u < 1$ for the bottom receiver, and 0 if left unmatched. The differentiation captures nonprice considerations, such as the receiver's reputation or other attributes of the match. The sender can match with at most one receiver. The labels top and bottom refer to the sender's preferences; the ranking of acceptance standards depends on the receivers' choices.

The certifier facilitates signaling through selling tests to the sender. The certifier seeks to maximize its revenue from selling tests to the sender. A test is defined by a set of test outcomes $\mathcal{M}_E$ and for each θ , a distribution $\rho_\theta \in \Delta(\mathcal{M}_E)$. When the sender takes a test, the outcomes are drawn from ρ_θ when the sender's quality is θ . The test outcome $m \in \mathcal{M}_E$ is revealed publicly. The certifier sets a (certification) *mechanism* Φ , which consists of a set of reports $\mathcal{M}$ and a function from elements of $\mathcal{M}$ to pairs of payments and tests. To allow for voluntary participation by the sender, I require that the "empty message" is always available to the sender at no cost.

Timing

1. The top and the bottom receivers simultaneously and publicly **commit** to acceptance standards $s = (s_t, s_b) \in [0, 1]^2$.
2. The certifier publicly announces a certification mechanism Φ .
3. The sender chooses a report in $\mathcal{M}$ and makes payment to the certifier as prescribed by Φ .

4. Receivers observe the certification mechanism and the test outcome $m \in \mathcal{M}_E$.¹⁰
5. The sender chooses a receiver among those whose posterior belief based on step 4 meets their acceptance standards, and remains unmatched otherwise.

The solution concept is *Perfect Bayesian Equilibrium*, henceforth referred to as just equilibrium.

Receivers commit to the expected quality of a sender that they are willing to accept (acceptance standards). This commitment assumption helps focus on receiver competition, avoiding considerations related to inter-market competition between the certifier and the receivers.¹¹

In a *direct* mechanism $\mathcal{M} = [0, 1]$ and $\mathcal{M}_E = A$. The space of test outcomes (or recommendations) is $A = \{a_t, a_b, r\} := \{\text{match with top receiver, match with bottom receiver, reject}\}$. In a direct mechanism, the test results specify whether to reject or accept the sender. Moreover, the recommendation to accept is receiver-specific. The sender reports his type $\mu \in [0, 1]$ and in response the certifier charges the sender a price $\varphi(\mu)$ and uses $\rho_\theta(\mu)$ to announce a recommendation $a \in A$.

A direct mechanism is *obedient* if it is optimal for each receiver to offer a match if and only if recommended to do so. More precisely, obedience entails that the receivers' beliefs are such that¹²

$$\Pr(\theta = h \mid a_t) \geq s_t > \Pr(\theta = h \mid a_b) \geq s_b > \Pr(\theta = h \mid r) \quad (1)$$

As test outcomes are public, obedience requires that the sender follow the certifier's recommendation. The strict inequalities in (1) follow as a sender (strictly) prefers the top receiver over the bottom receiver, and the bottom receiver over rejection.

Let $\rho_\theta^t(\mu) := \rho_\theta(\mu)(a_t)$ and $\rho_\theta^b(\mu) := \rho_\theta(\mu)(a_b)$, these represent the probability of matching with the top and bottom receivers respectively conditional on quality θ for type μ 's allocation. For a direct mechanism (ρ, φ) , we can represent a test allocation as $\rho = (\rho_h^t, \rho_h^b, \rho_l^t, \rho_l^b)$.

A direct mechanism yields *gross utility* for type μ equal to $V(\mu) = \mu (\rho_h^t(\mu) + u\rho_h^b(\mu)) + (1 - \mu) (\rho_l^t(\mu) + u\rho_l^b(\mu))$. Define the *indirect utility* function of a direct mechanism (ρ, φ) by

$$\mathcal{U}(\mu) := V(\mu) - \varphi(\mu)$$

¹⁰I require that both receivers can observe the test outcome. We can also allow the certifier to reveal test outcomes to subsets of receivers. The certifier does not gain from this extra contractual power, as on-path payoffs in revenue-maximizing equilibria remain unchanged. See obedience constraints in section 5 for details.

¹¹Without the commitment assumption, the certifier can manipulate the sender supply such that the expected quality of the sender, conditional on matching, is as low as possible ($= q$). Sequential rationality of the receiver then requires the receiver to accept the sender.

¹²For an incentive compatible direct mechanism (φ, ρ) , whenever $\int_0^1 \mu \rho_h(\mu)(a) dF(\mu) + \int_0^1 (1 - \mu) \rho_l(\mu)(a) dF(\mu) > 0$ Bayes rule implies

$$\Pr(\theta = h \mid a) := \frac{\int_0^1 \mu \rho_h(\mu)(a) dF(\mu)}{\int_0^1 \mu \rho_h(\mu)(a) dF(\mu) + \int_0^1 (1 - \mu) \rho_l(\mu)(a) dF(\mu)}$$

When $\int_0^1 \mu \rho_h(\mu)(a) dF(\mu) + \int_0^1 (1 - \mu) \rho_l(\mu)(a) dF(\mu) = 0$, the action a is never recommended and can be removed from (1).

When type μ misreports $v \neq \mu$ he earns gross utility given by

$$V(\mu, v) = \mu \left(\rho_h^t(v) + u \rho_h^b(v) \right) + (1 - \mu) \left(\rho_l^t(v) + u \rho_l^b(v) \right)$$

A direct mechanism is *incentive compatible* if for every μ, v

$$\mathcal{U}(\mu) \geq V(\mu, v) - \varphi(v).$$

A direct mechanism is *individually rational* if $\mathcal{U}(\mu) \geq 0$ for all μ .

Given acceptance standards (s_t, s_b) , the intermediary seeks to maximize revenue generated by selling tests to senders. By the revelation principle (Myerson (1986), Forges (1986)), it suffices to restrict attention to direct, obedient, and incentive compatible mechanisms (ρ, φ) with full participation. Thus, the certifier's objective is to maximize revenue

$$\Pi = \int_0^1 \varphi(\mu) dF(\mu)$$

among obedient, incentive compatible, individually rational direct mechanisms.

Moreover, by the definition of indirect utility for an obedient, incentive compatible, individually rational direct mechanism, we have the following

$$\Pi = \int_0^1 [V(\mu) - \mathcal{U}(\mu)] dF(\mu)$$

Given such a mechanism with test allocation ρ , the payoff to the receivers is

$$U_t(\rho) := \int_0^1 [\mu v_h \rho_h^t(\mu) + (1 - \mu) v_l \rho_l^t(\mu)] dF(\mu)$$

$$U_b(\rho) := \int_0^1 [\mu v_h \rho_h^b(\mu) + (1 - \mu) v_l \rho_l^b(\mu)] dF(\mu)$$

When the choice of test allocation is unambiguous, I will drop dependence on ρ from the notation of receiver payoffs.

4 No Screening Benchmark

I will first present the benchmark for the certifier's problem without screening frictions. In this section, the sender cannot misreport his type in a direct mechanism, and thus, all test allocations are incentive compatible. As the certifier faces no incentive compatibility restrictions, it can extract all surplus generated. This allocation is clearly individually rational. We are only left with obedience constraints, which constitute the certifier's information design problem. In particular, after observing the acceptance standards (s_t, s_b) , the optimal mechanism (ρ, φ) solves

$$\max_{\rho} \int_0^1 V(\mu) dF(\mu)$$

Such that ρ satisfies (1).

Thus, the design of optimal tests is as if the sender can commit ex-ante to a costless type-dependent test. I refer to this information design problem as the efficient benchmark, and the distortions described in section 6 are relative to this benchmark. Here, the notion of efficiency is for the sender-certifier market.

Proposition 1. *The benchmark admits equilibria in pure strategies with the following acceptance standards:*

$$\begin{cases} \{(\frac{1}{u}q, q)\} & \text{if } u > q \\ \{(s_t, s_b) = (1, x) \mid x \in [u, 1]\} & \text{otherwise} \end{cases}$$

Proof. See appendix A.1 □

Remark. In the absence of a competing receiver, the top receiver would always commit to accepting a sender only if it is certain that the sender's quality is $\theta = h$. In response, the certifier chooses a fully revealing test for all types. This gives the receiver the highest possible surplus as $v_h > 0 > v_l$. When there is a competing receiver, the top receiver faces the risk of losing out on potentially high quality senders to the bottom receiver. Thus, competition restricts the top receiver's selectivity.

The certifier maximizes the sender's gross utility subject to obedience (1). Allocating a sender to the top receiver yields a higher utility, but it comes at the cost of pooling fewer low quality senders. When the top receiver sets a higher standard than the bottom receiver, the certifier might find it beneficial to allocate senders to the bottom receiver. If s_b is sufficiently small, the lower marginal value of allocating to the bottom receiver is overshadowed by the greater mass of senders that the bottom receiver accepts. The interplay between these forces determines what tests are offered. When $s_t, s_b > \mathbb{E}_F[\mu]$, obedience binds for each receiver that matches with positive probability. If the top (bottom) receiver accepts with positive probability, then the expected quality of the matched sender equals the standard s_t (s_b). If a receiver's obedience constraint is slack, then the certifier can pool in more low quality senders to that receiver's recommendation, increasing the sender's gross utility. The trade-off described above can be seen by substituting the obedience constraints in the certifier's objective.

$$\mathbb{E} \left[\frac{\mu}{s_t} \rho_h^t(\mu) + u \frac{\mu}{s_b} \rho_h^b(\mu) \right]$$

The terms in front of ρ_h^t and ρ_h^b represent the sum of the direct value from test allocation conditional on high quality and the indirect value from pooling additional low quality senders. A simple pointwise maximization scheme then gives the solution. If $us_t < s_b$, the certifier only allocates senders to the top receiver. If $us_t > s_b$, the certifier only allocates the sender to the bottom receiver.

For $u \leq q$, the equilibria in [Proposition 1](#) give the top receiver its standalone allocation and standard $s_t = 1$.

If $q < u$, then the top receiver cannot set $s_t = 1$. Setting a high threshold gives the bottom receiver incentive to undercut the top receiver by choosing $s_b = u(1 - \varepsilon)$. More generally, given some s_t the bottom receiver can undercut the top receiver whenever there exists some $\varepsilon > 0$ such that $u(1 - \varepsilon)s_t > q$. In equilibrium, the receivers must not have an incentive to lower their standards to undercut the opponent. At $(s_t, s_b) = (q/u, q)$, neither receiver gains by undercutting the other. The certifier is indifferent between allocating to top and allocating to bottom; the response supporting this equilibrium allocates only to top.

5 Certification Design with Privately Informed Sender

I begin by describing how incentive compatibility affects the design of the certification mechanism. Consider some arbitrary but fixed acceptance standard (s_t, s_b) such that $s_t > s_b \geq q$.

Incentive Compatibility: Recall that a direct mechanism is incentive compatible if for every μ, ν

$$\mathcal{U}(\mu) \geq V(\mu, \nu) - \varphi(\nu).$$

An important quantity in the analysis of incentive compatibility is the *slope* of the indirect utility. Consider a function K defined by the difference in the sender's utility conditional on $\theta = h$ and the sender's expected utility conditional on $\theta = l$.

$$K(\mu) := \underbrace{\left(\rho_h^t(\mu) + u\rho_h^b(\mu) \right)}_{\text{utility conditional on } \theta=h} - \underbrace{\left(\rho_l^t(\mu) + u\rho_l^b(\mu) \right)}_{\text{utility conditional on } \theta=l}$$

As $V(\mu, \nu)$ is linear in the sender's type μ , by [Rochet \(1987\)](#) and [Chopra and Ely \(2025\)](#), a test allocation ρ is incentive compatible for some price φ if and only the following monotonicity condition holds

$$K(\mu) = \left(\rho_h^t(\mu) + u\rho_h^b(\mu) \right) - \left(\rho_l^t(\mu) + u\rho_l^b(\mu) \right) \text{ is non-decreasing}$$

Moreover, the function K describes the slope of the sender's indirect utility. In particular, $\mathcal{U}$ is convex and hence absolutely continuous. The indirect utility has the following integral representation

$$\mathcal{U}(\mu) = \mathcal{U}(0) + \int_0^\mu K(\nu) d\nu$$

Importantly, a test allocation ρ with lower variation in θ contingent outcomes results in a flatter indirect utility.

Obedience: Recall that the obedience constraint is given by

$$\Pr(\theta = h \mid a_t) \geq s_t > \Pr(\theta = h \mid a_b) \geq s_b > \Pr(\theta = h \mid r)$$

Revenue maximization requires that the top receiver's obedience constraint $\Pr(\theta = h \mid a_t) \geq s_t$ holds with equality. Assume for contradiction that ρ generates constraint $\Pr(\theta = h \mid a_t) > s_t$, then the certifier can improve its revenue by offering a lottery over test allocations. This lottery allocates all types with a small probability α , regardless of θ , to the top receiver, and with probability $(1 - \alpha)$ the improving lottery allocates according to ρ . This preserves obedience and incentive compatibility, but generates a greater sender gross utility without increasing information rents, thus increasing revenue. I present the argument for the binding bottom receiver obedience constraint in appendix A.2. The argument follows by splitting a small mass of senders allocated to the bottom receiver, conditional on $\theta = h$, and allocating them between the top receiver and rejection. Obedience constraints for a revenue maximizing mechanism can be expressed as follows¹³

$$\Pr(\theta = h \mid a_t) = s_t > \Pr(\theta = h \mid a_b) = s_b > \Pr(\theta = h \mid r) \quad (2)$$

The equality of obedience constraints is similar to the informed certifier benchmark (section 4), but the argument used to perturb suboptimal mechanisms explicitly accounts for the sender's incentive to misreport his type (incentive compatibility). From the definition of $\mathbb{E}[\mu|a]$, obedience constraint can be expressed by linear equalities, whenever $s_t > s_b \geq \mathbb{E}_F[\mu]$.¹⁴

$$(1 - s_t) \int_0^1 \mu \rho_h^t(\mu) dF(\mu) - s_t \int (1 - \mu) \rho_l^t(\mu) dF(\mu) = 0$$

$$(1 - s_b) \int_0^1 \mu \rho_h^b(\mu) dF(\mu) - s_b \int (1 - \mu) \rho_l^b(\mu) dF(\mu) = 0$$

Individual Rationality: Any incentive compatible and obedient direct mechanism is revenue maximizing only if the individual rationality constraint is binding for some type μ_0 . More precisely, there exists a type μ_0 such that $\mathcal{U}(\mu_0) = 0$. Let μ_0 be the smallest type with binding individual rationality. The indirect utility is then given by

$$\mathcal{U}(\mu) = \begin{cases} - \int_{\mu}^{\mu_0} K(v) dv, & \text{if } \mu \leq \mu_0 \\ \int_{\mu_0}^{\mu} K(v) dv, & \text{if } \mu \geq \mu_0 \end{cases}$$

¹³The argument for binding obedience constraints is independent of whether the certifier announces test outcomes publicly or not. Additionally, if $s_t \leq s_b$ then in any revenue-maximizing mechanism a_b is recommended with zero probability. Thus, it is without loss to assume public test outcomes.

¹⁴The equalities imply that the expected quality of a sender being recommended a_b is $s_b < s_t$, hence the sender does not deviate. As $s_t, s_b \geq \mathbb{E}[\mu]$, the obedience constraint for recommendation r follows from the law of total expectation.

The revenue can then be expressed as the difference between gross and indirect utility

$$\begin{aligned}\Pi &= \int_0^1 [V(\mu) - \mathcal{U}(\mu)] dF(\mu) \\ &= \int_0^{\mu_0} \left\{ V(\mu) + \int_{\mu}^{\mu_0} K(v) dv \right\} f(\mu) d\mu + \int_{\mu_0}^1 \left\{ V(\mu) - \int_{\mu_0}^{\mu} K(v) dv \right\} f(\mu) d\mu\end{aligned}$$

As $\mathcal{U}(\mu_0) = 0$, the slope of the indirect utility of all types lower than μ_0 is negative. Thus, the expression above increases when K increases for types $\mu \leq \mu_0$. Moreover, for $\mu < \mu_0$ either $\rho_l^t(\mu) > \rho_h^t(\mu)$ or $\rho_l^b(\mu) > \rho_h^b(\mu)$. Marginally increasing ρ_h^t or ρ_h^b for types just below μ_0 preserves incentive compatibility, and increases the revenue. But increasing ρ_h^t is not always feasible as it might lead to a decrease in ρ_h^b and a possible violation of the obedience constraint (1). This happens only if $\rho_h^t(\mu) + \rho_h^b(\mu) = 1$, which along with a negative slope implies $\rho_l^t(\mu) > 0$. Using this, Lemma 1 demonstrates how a simple perturbation can still be constructed to increase revenue.

Lemma 1. *Let (ρ, φ) be an incentive compatible, obedient, and individually rational direct mechanism with indirect utility $\mathcal{U}$. The mechanism (ρ, φ) is revenue maximizing only if $\mathcal{U}(0) = 0$.*

Proof. See appendix A.3 □

Lemma 1 implies that the revenue maximizing incentive compatible and obedient mechanism requires that the lowest type has a binding individual rationality constraint ($\mathcal{U}(0) = 0$). Importantly, this implies that the slope of the indirect utility is everywhere non-decreasing and nonnegative.

Distortions Relative to the Benchmark: Using integration by parts, we can then express the certifier's revenue from an optimal incentive compatible and obedient mechanism in the familiar virtual surplus representation, see Myerson (1981).

$$\mathbb{E} \left[\underbrace{(\rho_h^t(\mu) + u\rho_h^b(\mu)) \left(\mu - \frac{1 - F(\mu)}{f(\mu)} \right)}_{\text{high quality virtual value}} \right] + \mathbb{E} \left[\underbrace{(\rho_l^t(\mu) + u\rho_l^b(\mu)) \left(1 - \mu + \frac{1 - F(\mu)}{f(\mu)} \right)}_{\text{low quality virtual value}} \right]$$

The above expression reveals an interesting property of the certifier's problem; under standard monotone hazard rate conditions, see Myerson (1981), the high quality virtual value is increasing and crosses 0 from below, while the low quality virtual value is positive and decreasing. From the definition of K , we can express the objective as follows¹⁵

$$\mathbb{E} \left[\rho_h^t(\mu) + u\rho_h^b(\mu) \right] - \mathbb{E} \left[\left(1 - \mu + \frac{1 - F(\mu)}{f(\mu)} \right) K(\mu) \right]$$

¹⁵To keep the notation simple, I do not explicitly index K by the test choice ρ .

Plugging in the obedience constraint, which holds with equality, we can simplify the expression for the sender's gross utility. We can simplify the objective to the following

$$\underbrace{\mathbb{E} \left[\frac{\mu}{s_t} \rho_h^t(\mu) + u \frac{\mu}{s_b} \rho_h^b(\mu) \right]}_{\text{Information design tradeoff}} - \underbrace{\mathbb{E} \left[\left(\frac{1 - F(\mu)}{f(\mu)} \right) K(\mu) \right]}_{\text{Distortion from screening}} \quad (3)$$

The first term in (3) captures the information design tradeoff, studied in section 4. The second term is a distortionary term due to the certifier screening the sender's private information. Notice that the revenue generated by the optimal mechanism in the benchmark is greater than the revenue generated when the certifier has to screen the sender.

5.1 Certifier's Optimization Problem

Solving the certifier's primal problem involves aggregate linear constraints (in the form of obedience) and monotonicity constraints (for incentive compatibility). The formal optimization problem can be summarized as

$$\max_{\rho: [0,1] \rightarrow \Delta(A) \times \Delta(A)} \mathbb{E} [V(\mu)] - \mathbb{E} \left[\frac{1 - F(\mu)}{f(\mu)} K(\mu) \right] \quad (\text{P})$$

subject to

$$(1 - s_t) \int_0^1 \mu \rho_h^t(\mu) dF(\mu) - s_t \int (1 - \mu) \rho_l^t(\mu) dF(\mu) \geq 0 \quad (\text{Top Receiver Obedience})$$

$$(1 - s_b) \int_0^1 \mu \rho_h^b(\mu) dF(\mu) - s_b \int (1 - \mu) \rho_l^b(\mu) dF(\mu) \geq 0 \quad (\text{Bottom Receiver Obedience})$$

$$K \geq 0 \quad (\text{Individual Rationality})$$

$$K \text{ non-decreasing} \quad (\text{Incentive Compatibility})$$

Relaxed Problem: Consider the relaxation of the primal problem (P) which does not involve the monotonicity constraint for K .

$$\max_{\rho: [0,1] \rightarrow \Delta(A) \times \Delta(A)} \mathbb{E} [V(\mu)] - \mathbb{E} \left[\frac{1 - F(\mu)}{f(\mu)} K(\mu) \right] \quad (\text{R})$$

subject to

$$(1 - s_t) \int_0^1 \mu \rho_h^t(\mu) dF(\mu) - s_t \int (1 - \mu) \rho_l^t(\mu) dF(\mu) \geq 0$$

$$(1 - s_b) \int_0^1 \mu \rho_h^b(\mu) dF(\mu) - s_b \int (1 - \mu) \rho_l^b(\mu) dF(\mu) \geq 0$$

$$K \geq 0$$

In general, the solution to the relaxed problem (R) is not incentive compatible. To simplify the analysis and in the interest of focusing on economic insights, I require the prior cdf F to obey the following regularity assumption.

Assumption 1. The weighted inverse hazard ratio $\frac{1-F(\mu)}{(1-\mu)f(\mu)}$ is non-increasing on $(0, 1)$.¹⁶

Importantly, assumption 1 leads to the following key simplification:

Proposition 2. *If assumption 1 holds, then there is a solution to the relaxed problem (R) that is also a solution to the primal problem (P). Moreover under assumption 1, any solution to the primal problem (P) is also a solution of the relaxed problem (R).*

Proof. See appendix A.4 □

Remark. The proposition follows from rearranging an optimal allocation while preserving the high- and low-quality masses assigned to each receiver. If the incentive slope decreases between two sets of types, either high-quality gross utility decreases or low-quality gross utility increases between those sets. In the first case, shifting high-quality allocations toward higher types strictly increases revenue under Assumption 1. In the second, shifting low-quality allocations toward lower types weakly increases revenue. When the weighted inverse hazard ratio is flat, the latter change need not be a strict improvement. Appendix A.4 presents the high-quality allocation calculation.

5.2 Increased Affinity to Less Selective Receivers

The key theoretical insight here is that second-degree price discrimination by the certifier steers allocations towards less selective receivers. Less selective receivers accept senders with lower expected quality, allowing the certifier to offer tests with lower payoff variation (heterogeneity)

¹⁶Writing $g(\mu) = \frac{1-F(\mu)}{(1-\mu)f(\mu)}$ gives the representation

$$F(\mu) = 1 - \exp\left(-\int_0^\mu \frac{1}{(1-v)g(v)} dv\right).$$

The integral is interpreted as an improper integral when necessary. Continuity of f and the maintained condition $f(1) > 0$ imply

$$\lim_{\mu \uparrow 1} \frac{1-F(\mu)}{(1-\mu)f(\mu)} = 1.$$

Consequently, Assumption 1 implies the inverse-hazard lower bound

$$\frac{1-F(\mu)}{f(\mu)} \geq 1-\mu, \quad 0 < \mu < 1.$$

The endpoint condition is essential to this implication: a continuous density with full support alone does not ensure the displayed limit.

across sender types. The lower variation in test outcomes lowers information rents by flattening the sender's indirect utility. This highlights a unique wedge resulting from the interaction of screening and information design components of the certifier's problem. The certifier distorts the test allocation by reducing the overall informativeness of tests.

In the benchmark (section 4), if $us_t < s_b$ then the certifier allocates the sender only to the top receiver. The first statement of Theorem 1 shows a reversal in test allocation when the certifier has to screen the sender. For standards with $s_b - us_t$ sufficiently small (but positive), the certifier allocates the sender only to the bottom receiver. The second statement of Theorem 1 shows that for any choice of standards $s_t > s_b$, the certifier **never** chooses an allocation that allocates the sender only to the top receiver. Formally, we get the following result.

Theorem 1. *Under assumption 1, there exists a constant $\zeta > 0$ such that if $\zeta > s_b - us_t$ then the optimal incentive compatible, obedient and individually rational direct mechanism recommends a_t with zero probability ($\rho^t = 0$).*

Moreover, under assumption 1, if $s_t > s_b$ then the optimal incentive compatible, obedient and individually rational direct mechanism recommends a_b with positive probability ($\rho_h^b \neq 0$).

Proof. See appendix A.5 □

Remark. Theorem 1 highlights the underlying channel through which the screening frictions faced by the certifier shape receiver competition. In the benchmark in section 4, the certifier only allocates the sender to the bottom receiver if $us_t > s_b$. When the bottom receiver sets a lower standard than the top receiver, the certifier can pool a greater mass of low quality senders if it allocates the sender to the bottom receiver over the top receiver. The distortions from the screening by the certifier amplify this effect. To observe this, recall the certifier's revenue is given by (3)

$$\mathbb{E} \left[\frac{\mu}{s_t} \rho_h^t(\mu) + u \frac{\mu}{s_b} \rho_h^b(\mu) \right] - \mathbb{E} \left[\frac{1 - F(\mu)}{f(\mu)} K(\mu) \right]$$

The slope $K(\mu)$ is given by the difference in the sender's utility conditional on $\theta = h$ and the sender's utility conditional on $\theta = l$.

$$K(\mu) = \left(\rho_h^t(\mu) + u \rho_h^b(\mu) \right) - \left(\rho_l^t(\mu) + u \rho_l^b(\mu) \right)$$

A reduction in quality contingent variation (heterogeneity) of test outcomes decreases the quality contingent variation in the sender's utility, resulting in lower information rents. A larger mass of low quality senders reduces the difference in the test outcome distribution conditional on θ . Due to this additional benefit, the certifier "overallocates" the sender to the less selective receiver, even if it reduces the sender's gross utility. The resulting allocation can favor the bottom receiver even at standards where the benchmark allocates only to top.

6 Equilibrium Analysis

Having described the certifier's response to standards chosen by the receivers, we are ready to describe the equilibrium. The feedback between the certification mechanism and acceptance standards determines how credential rationing affects the allocation of senders across receivers.

Equilibrium Well-posedness: In general, for any given pair of standards (s_t, s_b) , there can be many optimal mechanisms. More precisely, the set of solutions to (P) might not be a singleton. Thus, the certifier's sequential rationality alone does not pin down the certification mechanism. This is particularly troublesome when the payoffs of the top and the bottom receivers vary across different solutions.¹⁷ Conveniently for any given (s_t, s_b) , under assumption 1, the payoffs of the receivers are constant over the set of optimal mechanisms. Moreover, the induced receiver payoffs vary continuously with (s_t, s_b) . This establishes that under assumption 1, an equilibrium of the certification game exists with receivers playing possibly mixed strategies in the first period. Lemma 2 formalizes these claims

Lemma 2. *Let F satisfy assumption 1. For each $(s_t, s_b) \in [q, 1]^2$, the receiver payoffs U_t and U_b are constant over the set of solutions to the primal problem (P). Moreover, the payoffs U_t and U_b vary continuously with (s_t, s_b) .*

Proof. See appendix A.7 □

Let $\sigma = (\sigma_t, \sigma_b) \in \Delta([0, 1]) \times \Delta([0, 1])$ represent a mixed strategy profile played by the receivers.

6.1 Single Receiver

To demonstrate the equilibrium interaction of acceptance standards and certification mechanisms, I will consider a slight variation of the model with only the top receiver, or equivalently $u = 0$. Given a standard $s_t > \mathbb{E}[\mu]$, the obedience constraint is

$$(1 - s_t)\mathbb{E}[\mu\rho_h^t(\mu)] - s_t\mathbb{E}[(1 - \mu)\rho_l^t(\mu)] \geq 0$$

Writing $\rho_l^t = \rho_h^t - K$ and rearranging yields

$$\mathbb{E}\left[\left(1 - \frac{\mu}{s_t}\right)\rho_h^t(\mu)\right] \leq \mathbb{E}[(1 - \mu)K(\mu)]$$

By equation 3, we get the certifier's objective

$$\mathbb{E}\left[\frac{\mu}{s_t}\rho_h^t(\mu)\right] - \mathbb{E}\left[\frac{1 - F(\mu)}{f(\mu)}K(\mu)\right]$$

¹⁷In this case, the description of an equilibrium will involve the receiver's conjectures about the certifier's response to a deviation from equilibrium acceptance standards. This might be dependent on ad hoc tie-breaking assumptions.

Example: Assume that the distribution F is uniform, which implies $\frac{1-F(\mu)}{f(\mu)} = 1 - \mu$. The revenue expression simplifies further by plugging in the definition of K and using the binding obedience constraint to substitute $\mathbb{E} \left[\frac{1-F(\mu)}{f(\mu)} K(\mu) \right]$ with $\mathbb{E} \left[\left(1 - \frac{\mu}{s_t}\right) \rho_h^t(\mu) \right]$ and is given by

$$\mathbb{E} \left[\left(\frac{2\mu}{s_t} - 1 \right) \rho_h^t(\mu) \right]$$

Pointwise maximizing the integrand gives that $\rho_h^t(\mu) = 1$ for all $\mu \geq s_t/2$ and 0 otherwise. The corresponding ρ_l^t is such that $\rho_l^t(\mu) = \alpha$ if $\mu \geq s_t/2$ and 0 otherwise, where α solves

$$\alpha s_t \int_{s_t/2}^1 (1 - \mu) d\mu = (1 - s_t) \int_{s_t/2}^1 \mu d\mu$$

The pointwise optimal ρ_h^t does not satisfy the obedience constraint for $s_t < 2/3$, as the corresponding $\alpha > 1$. To restore obedience, the optimal solution involves allocating types below $s_t/2$ to the top receiver. When $s_t < 2/3$, the optimal mechanism is a bang-bang test. All types below a threshold $\mu_0 < s_t/2$ are excluded (allocated to the top receiver with zero probability) and all types above μ_0 are allocated to the top receiver with probability 1, independent of θ . Obedience constraint then implies that μ_0 solves

$$s_t \int_{\mu_0}^1 (1 - \mu) d\mu = (1 - s_t) \int_{\mu_0}^1 \mu d\mu$$

Recall the top receiver's payoff from test allocation ρ is

$$U_t = \mathbb{E} [\mu v_h \rho_h^t(\mu) + (1 - \mu) v_l \rho_l^t(\mu)]$$

By binding obedience constraint for acceptance standard s_t

$$U_t = \left(v_h + v_l \frac{1 - s_t}{s_t} \right) \mathbb{E} [\mu \rho_h^t(\mu)]$$

Given the designer's optimal response to $s_t \geq 2/3$, the top receiver's payoff is

$$\left(v_h + v_l \frac{1 - s_t}{s_t} \right) \int_{s_t/2}^1 \mu d\mu$$

This expression is increasing in s_t whenever $-v_h \geq v_l$, this holds as $q > 1/2$. Thus, the optimal acceptance standard is $s_t = 1$. When $s_t < 2/3$ the allocation takes the bang-bang form mentioned before. As $1/2 = \mathbb{E}[\mu] < q \leq s_t$, the threshold $\mu_0 < s_t/2 \leq 1/3$. Moreover, the binding obedience constraint implies that the designer responds to an increase in s_t by increasing threshold μ_0 . As $\mu_0 < 1/3 < q$, the expected quality of senders that are excluded from the mechanism after marginally increasing s_t is below q . Hence, when $s_t < 2/3$ the top receiver has a profitable deviation by increasing its acceptance standards. Generalizing this example we have Proposition

3

Proposition 3. Fix $u = 0$. If $\frac{1-F(\mu)}{f(\mu)}$ is convex and assumption 1 holds, then the top receiver's payoff U_t is increasing in s_t . In particular, the equilibrium standard is such that $s_t = 1$ with probability 1.

Proof. See appendix A.6 □

Remark. The above example illustrates how the sender's private information distorts equilibrium acceptance standards. Unlike the benchmark in section 4, the presence of private information leads to *exclusion*. To reduce information rents, the designer excludes lower types from the mechanism. Increasing standards results in more exclusion. Thus, the receiver faces a tradeoff: higher standards improve the expected quality of each match, but reduce the matching probability. In the uniform example, this trade off is resolved in the top receiver raising its standards.

Generally, the designer can respond to increased standards in one of two possible ways. First, increasing the probability with which the sender gets matched. The expected quality of these new senders must be greater than the receiver's standard, and hence greater than q . In particular, this leads to a greater payoff for the receiver. Second, by decreasing the probability with which the sender gets matched. The expected quality of the removed sender types must be less than the receiver's standard. But the expected quality of removed senders could still be greater than q , in which case the receiver's payoff decreases. In the example, the designer chooses the latter of the two responses. Yet, the receiver finds it profitable to increase standards as the expected quality of removed sender types is lower than q .

If $\frac{1-F(\mu)}{f(\mu)}$ is convex and $\frac{1-F(\mu)}{(1-\mu)f(\mu)}$ is non-increasing, then the designer responds by decreasing the mass of matched senders. But the expected quality of these senders is below $\mathbb{E}[\mu]$. Hence, the receiver sets $s_t = 1$.

6.2 Receiver Competition

I measure receiver competition by the receivers' shares of matches. Let $\sigma = (\sigma_t, \sigma_b)$ be the receivers' strategies in an equilibrium, and let ρ be the certifier's response. Given standards $s = (s_t, s_b)$, the total mass allocated to receiver j is

$$P_j(s, \rho) = \int_0^1 \left[\mu \rho_h^j(\mu) + (1 - \mu) \rho_l^j(\mu) \right] dF(\mu), \quad j \in \{t, b\}.$$

Write $\bar{P}_j(\sigma) = \mathbb{E}_\sigma[P_j(s, \rho)]$ for its expected matching mass and $\bar{U}_j(\sigma) = \mathbb{E}_\sigma[U_j(s, \rho)]$ for its expected payoff. Its share of all matches is

$$M_j(\sigma) = \frac{\bar{P}_j(\sigma)}{\bar{P}_t(\sigma) + \bar{P}_b(\sigma)}. \quad (4)$$

Thus $M_j(\sigma)$ is the probability of matching with receiver j , conditional on a match occurring.

Theorem 2. Suppose Assumption 1 holds and $\mathbb{E}_F[\mu] < q < 1$. Let σ be an equilibrium with screening, and let σ^0 be a benchmark equilibrium identified in Proposition 1. In equilibrium σ , both receivers earn strictly positive expected payoffs, and

$$M_t(\sigma^0) = 1 > M_t(\sigma) > 0, \quad M_b(\sigma^0) = 0 < M_b(\sigma) < 1.$$

Proof. See Appendix A.8. □

Theorem 2 shows a peculiar spillover effect that allocative inefficiencies have on receiver competition. The theorem is driven by two central features of the certification market. First, the nature of inefficiencies resulting from the interaction of screening and information design (Theorem 1). Second, the acceptance standards shape the certifier's information design problem, resulting in the equilibrium dependence of the certification mechanism and receivers' acceptance standards. Screening gives the bottom receiver a positive share of matches and a positive expected payoff. The top receiver's share consequently falls relative to the benchmark allocations in Proposition 1.

7 Extensions

7.1 Wages

Consider the labor-market application of the base model, with an applicant as the sender and employers as the receivers. Instead of setting hiring standards, the employers now compete over wages. The employers simultaneously announce wages. In response, the certifier designs a mechanism. The applicant is hired if his expected value to the employer, conditional on test outcomes, is greater than the wage offered. To model imperfect competition through vertical differentiation, I assume that w dollars from the top employer are worth $\gamma_t(w) > w$ to the applicant and w dollars from the bottom employer are worth w to the applicant. Like before, we can apply the revelation principle to focus on incentive compatible, individually rational, and obedient direct mechanisms. Given wages $(w_t, w_b) \in [0, v_h]^2$, the obedience constraints for the employers are

$$\mathbb{E}[v_\theta \mid a_t] \geq w_t$$

and

$$\mathbb{E}[v_\theta \mid a_b] \geq w_b$$

Conditional on hiring, an employer's profit is the expected productivity of the applicant minus the wage.

The applicant's obedience requires that he join the employer that is recommended by the certifier. Type μ applicant's gross utility from reporting v

$$V_w(\mu, v) = \mu \left(\gamma_t(w_t) \rho_h^t(v) + w_b \rho_h^b(v) \right) + (1 - \mu) \left(\gamma_t(w_t) \rho_l^t(v) + w_b \rho_l^b(v) \right)$$

This reduces to the base model in section 3 by defining

$$s_t = \frac{w_t - v_l}{v_h - v_l}, \quad s_b = \frac{w_b - v_l}{v_h - v_l}, \quad \text{and} \quad u = \frac{w_b}{\gamma_t(w_t)}$$

We can express the applicant's gross utility as

$$V_w(\mu, v) = \gamma_t(w_t) \times V(\mu, v)$$

The results from section 5 can be applied to this setting. In particular, the obedience constraint for the employers is binding. The expected value of the applicant, conditional on hiring, equals the offered wage, leading to zero profit for employers. The certifier's ability to flexibly design information allows it to freely pool low ability applicants, reducing the employers' profits. In fact, the model predicts that vertical differentiation of employers alone does not lead to meaningful employer competition when the certifier can flexibly design and price information.

7.2 Two Sided Platform

Consider an extension of the base model in which the certifier can charge the receivers for participating in the mechanism. This may arise in two-sided markets where the certifier charges both the sender and the receivers. To fix ideas, I model the fee as a fixed fraction of the receivers' profits from the sender match. Let $\alpha = (\alpha_t, \alpha_b) \in [0, 1]^2$ represent the fraction of receivers' surplus that the certifier charges the receivers for using its services. Like before, we can restrict attention to incentive compatible, individually rational, and obedient direct mechanisms. The analysis of the sender's and receivers' problems is mostly unchanged. The certifier's revenue from a test allocation ρ is given by

$$\alpha_t U_t(\rho) + \alpha_b U_b(\rho) + \mathbb{E} [V(\mu) - \mathcal{U}(\mu)]$$

The above expression simplifies to

$$\mathbb{E} \left[\mu \left(x_h^t \rho_h^t(\mu) + x_h^b \rho_h^b(\mu) \right) + (1 - \mu) \left(x_l^t \rho_l^t(\mu) + x_l^b \rho_l^b(\mu) \right) \right] - \mathbb{E} [\mathcal{U}(\mu)]$$

Here $x_\theta^t = 1 + \alpha_t v_\theta$ and $x_\theta^b = u + \alpha_b v_\theta$.

From the expression above, we can see that Proposition 2 extends to this setting. When the certifier can extract surplus from the receivers, the test allocations might be more informative. The fee charged to the receivers aligns the certifier's preference with the receivers'. The distortion from screening will still affect the equilibrium acceptance standards, but the effect is now confounded by the certifier's incentive to extract surplus from the receivers. Depending on the weights α_t and α_b , certification could be skewed in favor of either of the receivers.

7.3 Beyond Binary Quality

Receivers often value senders for many match-relevant characteristics. This becomes especially interesting if receivers value different characteristics differently. Putting aside the question of distortion to receiver competition, designing an optimal mechanism in this case is a multi-dimensional screening problem. This poses new challenges in the design of an optimal mechanism and the subsequent equilibrium analysis. A first step towards this should involve the special case when all receivers have similar preferences for different characteristics of a sender. I don't explore this possibility, but I conjecture that the link between competition and senders' private information remains in this special case.

8 Conclusion

The paper identifies a novel channel between credential rationing and competition between receivers. Second-degree price discrimination by the certifier can increase the bottom receiver's share of matches and reduce the top receiver's share relative to the no-screening benchmark.

Therefore, interventions in the certifier-sender market might have unexpected consequences for receiver competition. The model can also describe a borrower seeking certification to obtain financing, or a seller seeking quality certification to trade with a buyer, when the counterparties' terms are fixed and certification affects their willingness to match. These interpretations preserve the same sender-receiver structure. Moreover, equilibrium feedback between the decision makers' (receivers') actions and the certification mechanism highlights potential regulatory challenges in information markets more generally (see [Bergemann and Bonatti \(2019\)](#) for a survey of markets for information).

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A Appendix

A.1 Benchmark

Proposition 1. The benchmark admits equilibria in pure strategies with the following acceptance standards:

$$(s_t, s_b) = \begin{cases} (\frac{1}{u}q, q) & \text{if } u > q \\ (1, x) & \text{for } x \in [u, 1] \text{ otherwise} \end{cases}$$

Proof. Note that in any equilibrium, the receiver accepts a sender only if the sender's expected value is positive. Thus, q is a lower bound for the equilibrium standard set by either receiver. Moreover, in any equilibrium $s_t \geq s_b$, as otherwise the top receiver has a profitable deviation to $s_t = s_b$. The designer's problem can be stated as

$$\max \mathbb{E} \left[\mu (\rho_h^t(\mu) + u\rho_h^b(\mu)) + (1 - \mu)(\rho_l^t(\mu) + u\rho_l^b(\mu)) \right]$$

subject to

$$\Pr(\theta = h \mid a_t) \geq s_t > \Pr(\theta = h \mid a_b) \geq s_b > \Pr(\theta = h \mid r)$$

As $s_t, s_b \geq q > \mathbb{E}[\mu]$, the obedience constraints are binding for a revenue maximizing testing policy. If the obedience constraints are slack then the certifier can do better by allocating more low quality senders to the receiver with slack obedience constraint. Plugging the binding obedience constraints into the expression for the designer's revenue yields

$$\mathbb{E} \left[\frac{\mu}{s_t} \rho_h^t(\mu) + u \frac{\mu}{s_b} \rho_h^b(\mu) \right]$$

The optimal test allocation is then

$$(\rho_h^t, \rho_h^b) = \begin{cases} (1, 0) & \text{if } us_t < s_b \\ (0, 1) & \text{if } us_t > s_b \\ (\alpha, 1 - \alpha) & \text{for some } \alpha \in [0, 1] \text{ otherwise} \end{cases}$$

Note under binding obedience constraint the receiver's payoff can be expressed as:

$$U_t(\rho) = \left(v_h + v_l \frac{1 - s_t}{s_t} \right) \mathbb{E} [\mu \rho_h^t(\mu)]$$

$$U_b(\rho) = \left(v_h + v_l \frac{1 - s_b}{s_b} \right) \mathbb{E} [\mu \rho_h^b(\mu)]$$

If $s_t < \max(1, s_b/u)$ then the top receiver has a profitable deviation by increasing standards. If $us_t > s_b$ and $us_t > q$ then the bottom receiver prefers to increase its standard to $s_b = us_t - \varepsilon$ for small enough $\varepsilon > 0$.

Finally, in equilibrium, the bottom receiver should be unable to undercut the top receiver. Thus, $us_t \leq q$. Moreover, the equilibrium test allocation has $(\rho_h^t, \rho_h^b) = (1, 0)$ as otherwise the top receiver can undercut the bottom receiver. This establishes the Proposition. $\square$

A.2 Obedience

Consider ρ such that $\Pr(\theta = h \mid a_t) = s_t > \Pr(\theta = h \mid a_b) > s_b$ and a_b is recommended with positive probability. There must be some positive probability set S of types such that $\rho_h^b(\mu) > 0$ for all $\mu \in S$. For any positive probability subset $\hat{S} \subset S$, the designer can perturb the test allocation as described below. For all $\mu \in \hat{S}$, conditional on $\theta = h$, the new test allocates the sender to the top receiver with probability $\rho_h^t(\mu) + u\rho_h^b(\mu)$ and rejects with probability $1 - \rho_h^t(\mu) - u\rho_h^b(\mu)$. Otherwise, the test allocation is unchanged. The new test neither changes the gross utility nor the indirect utility, yielding the same revenue as ρ . This results in a slack obedience constraint for the top receiver. The bottom receiver's obedience constraint is not violated as the perturbation can be made over arbitrary subsets $\hat{S}$. Combining this with the argument in the main text implies that ρ is not revenue maximizing. Thus, for optimal mechanisms, the obedience constraint holds with equality.

A.3 Individual Rationality

Lemma 1. Let (ρ, φ) be an incentive compatible, obedient, and individually rational direct mechanism with indirect utility $\mathcal{U}$. The mechanism (ρ, φ) is revenue maximizing only if $\mathcal{U}(0) = 0$.

Proof. Consider (ρ, φ) such that the binding type is $\mu_0 > 0$. From the arguments in the main text, we get that there is some non-increasing function $\gamma : [0, 1] \rightarrow [0, 1]$ and $\varepsilon > 0$ such that $\gamma(\mu_0) = 0$, $\gamma(\mu) > 0$ for some $\mu \in [\mu_0 - \varepsilon, \mu_0]$ and the following perturbed test allocation $\hat{\rho}$ is feasible and incentive compatible

$$\begin{aligned} (\hat{\rho}_h^t, \hat{\rho}_h^b) &= \begin{cases} (\rho_h^t(\mu) + \gamma(\mu), \rho_h^b(\mu)) & \text{if } \mu_0 - \varepsilon \leq \mu < \mu_0 \text{ and } \rho_h^t(\mu) + \rho_h^b(\mu) < 1 \\ (\rho_h^t(\mu), \rho_h^b(\mu)) & \text{otherwise} \end{cases} \\ (\hat{\rho}_l^t, \hat{\rho}_l^b) &= \begin{cases} (\rho_l^t(\mu) - \gamma(\mu), \rho_l^b(\mu)) & \text{if } \mu_0 - \varepsilon \leq \mu < \mu_0 \text{ and } \rho_h^t(\mu) + \rho_h^b(\mu) = 1 \\ (\rho_l^t(\mu), \rho_l^b(\mu)) & \text{otherwise} \end{cases} \end{aligned}$$

Let $S, S' \subset [\mu_0 - \varepsilon, \mu_0]$ such that $\rho_h^t(\mu) + \rho_h^b(\mu) < 1$ on S and $\rho_h^t(\mu) + \rho_h^b(\mu) = 1$ on S' . The change in revenue from this perturbation is

$$\begin{aligned} & \int_0^{\mu_0} \left\{ \hat{V}(\mu) - V(\mu) + \int_{\mu}^{\mu_0} [\hat{K}(v) - K(v)] dv \right\} f(\mu) d\mu \\ &= \int_0^{\mu_0 - \varepsilon} \left[\int_{\mu_0 - \varepsilon}^{\mu_0} \gamma(v) dv \right] f(\mu) d\mu \\ &+ \int_{\mu_0 - \varepsilon}^{\mu_0} \left[\int_{\mu}^{\mu_0} \gamma(v) dv \right] f(\mu) d\mu \\ &+ \int_S \mu \gamma(\mu) f(\mu) d\mu - \int_{S'} (1 - \mu) \gamma(\mu) f(\mu) d\mu \end{aligned}$$

$$\begin{aligned}
&= F(\mu_0 - \varepsilon) \int_{\mu_0 - \varepsilon}^{\mu_0} \gamma(\mu) d\mu \\
&\quad + \int_{\mu_0 - \varepsilon}^{\mu_0} [F(\mu) - F(\mu_0 - \varepsilon)] \gamma(\mu) d\mu + \int_S \mu \gamma(\mu) f(\mu) d\mu - \int_{S'} (1 - \mu) \gamma(\mu) f(\mu) d\mu \\
&= \int_{\mu_0 - \varepsilon}^{\mu_0} \gamma(\mu) F(\mu) d\mu + \int_S \mu \gamma(\mu) f(\mu) d\mu - \int_{S'} (1 - \mu) \gamma(\mu) f(\mu) d\mu \\
&\geq \int_{\mu_0 - \varepsilon}^{\mu_0} \gamma(\mu) F(\mu) d\mu - \int_{\mu_0 - \varepsilon}^{\mu_0} (1 - \mu) \gamma(\mu) f(\mu) d\mu
\end{aligned}$$

Notice that the change in revenue is composed of positive terms except for the reduction in the sender's surplus from a decreased matching probability of low quality senders in the set S' . When $\int_{\mu_0 - \varepsilon}^{\mu_0} \gamma(\mu) F(\mu) d\mu + \int_S \mu \gamma(\mu) f(\mu) d\mu - \int_{S'} (1 - \mu) \gamma(\mu) f(\mu) d\mu < 0$ we can further perturb the test by offering a lottery of test $\hat{\rho}$ and the test $(1, 0, 1, 0)$. If ρ is obedient, then $\hat{\rho}$ has slack obedience constraint for the top receiver. In particular, we can further perturb the allocation into $\rho' = (1 - \alpha)\hat{\rho} + \alpha(1, 0, 1, 0)$. Where $\alpha \in (0, 1)$ is chosen such that

$$\alpha s_t \int_0^1 (1 - \mu) dF(\mu) - (1 - \alpha) s_t \int_{\mu_0 - \varepsilon}^{\mu_0} (1 - \mu) \gamma(\mu) f(\mu) d\mu = \alpha (1 - s_t) \int_0^1 \mu dF(\mu)$$

Rearranging the terms, we get the following

$$\alpha \left(1 - \frac{\mathbb{E}[\mu]}{s_t} \right) = (1 - \alpha) \int_{\mu_0 - \varepsilon}^{\mu_0} (1 - \mu) \gamma(\mu) f(\mu) d\mu$$

As ε can be made arbitrarily small, such an α exists in the interval $(0, 1)$. By the choice of α , the perturbed test ρ' is obedient.

For $s_b \geq us_t$ we get that $\Pi(\rho) \leq \frac{\mathbb{E}[\mu]}{s_t}$, as $\frac{\mathbb{E}[\mu]}{s_t}$ is the revenue generated by the optimal mechanism from the benchmark in section 4. In particular, we get the following

$$\alpha (1 - \Pi(\rho)) \geq (1 - \alpha) \int_{\mu_0 - \varepsilon}^{\mu_0} (1 - \mu) \gamma(\mu) f(\mu) d\mu \geq (1 - \alpha) (\Pi(\rho) - \Pi(\hat{\rho}))$$

This implies

$$\Pi(\rho') - \Pi(\rho) > 0$$

Thus, we can choose some $\alpha \in (0, 1)$ such that the test ρ' is incentive compatible, obedient, and yields a greater revenue than ρ .

Whenever $s_b < us_t$ the optimal mechanism only allocates the sender to the bottom receiver or to being unmatched (bottom receiver-only allocation). If $s_b < us_t$ then the bottom receiver-only allocation can produce at least as much gross surplus as any other allocation. Additionally, as $s_b < s_t$, the certifier can pool a greater mass of low-quality senders. Thus, a bottom receiver-only allocation minimizes the information rent by reducing $K(\mu) = (\rho_h^t(\mu) + u\rho_h^b(\mu)) - (\rho_l^t(\mu) + u\rho_l^b(\mu))$ while keeping the sender's gross surplus fixed. In this case, the argument in text shows

that $\mathcal{U}(0) = 0$ as for a bottom receiver-only allocation negative slope $K(\mu) < 0$ implies that $\rho_h^t(\mu) + \rho_h^b(\mu) < 1$. □

A.4 Solution to Relaxed Problem

Geometry of Optimal Tests: For the formal analysis, consider ρ as an element of the Hilbert space $L_2([0, 1] \rightarrow \mathbb{R}^4, F)$ with the usual norm topology and Borel sigma algebra. Where $\rho = (\rho_h^t, \rho_h^b, \rho_l^t, \rho_l^b)$. I maintain this assumption for the rest of the appendix.

For the relaxed problem, we can describe a test allocation ρ as a function from $[0, 1]$ to the closed convex polytope $\mathcal{C}$. Where

$$\mathcal{C} = \left\{ x \in \mathbb{R}_+^4 \mid \begin{array}{l} x_1 + x_2 \leq 1 \\ x_3 + x_4 \leq 1 \\ x_1 + ux_2 - x_3 - ux_4 \geq 0 \end{array} \right\}$$

Recall, the relaxed problem (R) is given by the following

$$\max_{\rho: [0,1] \rightarrow \Delta(A) \times \Delta(A)} \mathbb{E} \left[\mu(\rho_h^t(\mu) + u\rho_h^b(\mu)) + (1 - \mu)(\rho_l^t(\mu) + u\rho_l^b(\mu)) \right] - \mathbb{E} \left[\left(\frac{1 - F(\mu)}{f(\mu)} \right) K(\mu) \right]$$

subject to

$$\begin{aligned} (1 - s_t) \int_0^1 \mu \rho_h^t(\mu) dF(\mu) - s_t \int_0^1 (1 - \mu) \rho_l^t(\mu) dF(\mu) &\geq 0 \\ (1 - s_b) \int_0^1 \mu \rho_h^b(\mu) dF(\mu) - s_b \int_0^1 (1 - \mu) \rho_l^b(\mu) dF(\mu) &\geq 0 \\ K &\geq 0 \end{aligned}$$

Define the feasibility set

$$\mathcal{F} := \left\{ \rho \in L_2([0, 1] \rightarrow [0, 1]^4) \mid \rho_h^t(\mu) + \rho_h^b(\mu) \leq 1, \rho_l^t(\mu) + \rho_l^b(\mu) \leq 1, K(\mu; \rho) \geq 0 \ \forall \mu \in [0, 1] \right\}$$

Here $\rho = (\rho_h^t, \rho_h^b, \rho_l^t, \rho_l^b)$. The set $\mathcal{F}$ is bounded, convex, and norm-closed.¹⁸ As L_2 is a reflexive Banach space, $\mathcal{F}$ is weakly closed and hence weakly compact.¹⁹ The objective function is a continuous linear functional of ρ , thus a solution to the relaxed problem exists in $\mathcal{F}$. The obedience constraints satisfy the usual Robinson constraint quantification, see [Robinson \(1975\)](#) and ch 8 of [Luenberger \(1997\)](#).²⁰ Thus for each solution ρ^* of (R) there exists a multiplier $\lambda^* = (\lambda_t^*, \lambda_b^*) \in \mathbb{R}_+^2$

¹⁸Where norm closedness follows from the fact that convergence in L_2 norm implies pointwise convergence almost everywhere over a subsequence.

¹⁹See chapter 2 of [Bonnans and Shapiro \(2013\)](#) for details.

²⁰This amounts to having non-empty interior of the set of all obedient test allocations in $\mathcal{F}$.

such that the corresponding Lagrangian $\mathcal{L}$ has a saddle point at (ρ^*, λ^*) . More precisely, for all $\rho \in \mathcal{F}$ and $\lambda \in \mathbb{R}_+^2$, the following holds

$$\mathcal{L}(\rho, \lambda^*) \leq \mathcal{L}(\rho^*, \lambda^*) \leq \mathcal{L}(\rho^*, \lambda)$$

Where the Lagrangian is given by

$$\mathcal{L}(\rho, \lambda) := \int_0^1 \langle l(\mu, \lambda), \rho(\mu) \rangle dF(\mu)$$

Where $\langle \cdot, \cdot \rangle$ is the usual inner product in $\mathbb{R}^4$ and

$$\begin{aligned} l(\mu, \lambda) := & \left(\mu - \frac{1 - F(\mu)}{f(\mu)}, u \left(\mu - \frac{1 - F(\mu)}{f(\mu)} \right), 1 - \mu + \frac{1 - F(\mu)}{f(\mu)}, u \left(1 - \mu + \frac{1 - F(\mu)}{f(\mu)} \right) \right) \\ & + (\mu \lambda_t (1 - s_t), \mu \lambda_b (1 - s_b), -(1 - \mu) \lambda_t s_t, -(1 - \mu) \lambda_b s_b) \end{aligned}$$

By the saddle point property above, we get

$$\begin{aligned} \mathcal{L}(\rho^*, \lambda^*) &= \max_{\rho \in \mathcal{F}} \int_0^1 \langle l(\mu, \lambda^*), \rho(\mu) \rangle dF(\mu) \\ &= \int_0^1 \max_{\rho(\mu) \in \mathcal{C}} \langle l(\mu, \lambda^*), \rho(\mu) \rangle dF(\mu) \end{aligned}$$

By linearity of inner product and the fact that linear functions optimized over closed and convex polytopes achieve optima at an extreme point, we get²¹

$$\rho^*(\mu) \in \text{ex}(\mathcal{C}) \quad \text{for all } \mu$$

The set $\text{ex}(\mathcal{C})$ can be enumerated as²²

$$\text{ex}(\mathcal{C}) = \left\{ \begin{array}{l} (0, 0, 0, 0) \\ (0, 1, 0, 1), (u, 0, 0, 1), (0, 1, u, 0), (1, 0, 1, 0) \\ (1, 0, 0, 1), (0, 1, 0, 0) \\ (1, 0, 0, 0) \end{array} \right\}$$

In particular, the optimal test allocation ρ^* takes finitely many values. This establishes the key assumption made in the proof of Proposition 2 sketched in section 5.1 about the optimal test taking finitely many values.

²¹For a closed convex polytope X the set $\text{ex}(X)$ is the set of all extreme points of X .

²²Note, the representation of $\text{ex}(\mathcal{C})$ is to emphasis monotonicity of K .

A.4.1 Proof of Proposition 2

To prove Proposition 2, extend the argument in Section 5.1 to arbitrary ordered sets $S_1 < S_2$ and their subsets S'_1, S'_2 . To complete the proof, I present the high-quality allocation calculation.

Fix some test allocation ρ . Consider two subsets S_1 and S_2 of $[0, 1]$ such that $S_1 < S_2$ in the strong set order. Let the slope K be constant on these sets. Assume for contradiction that $K(S_1) > K(S_2)$ and that

$$\rho_h^t(S_2) + u\rho_h^b(S_2) < \rho_h^t(S_1) + u\rho_h^b(S_1)$$

Consider subsets $S'_1 \subset S_1$ and $S'_2 \subset S_2$ such that

$$\int_{S'_1} \mu dF(\mu) = \int_{S'_2} \mu dF(\mu) > 0 \quad (\text{MC2})$$

We can perturb ρ_h on the sets S'_1 and S'_2 . Define, for $j \in \{t, b\}$ and $\varepsilon > 0$, the following perturbations

$$\begin{aligned} \tilde{\rho}_h^j(S'_1) &:= (1 - \varepsilon)\rho_h^j(S_1) + \varepsilon\rho_h^j(S_2) \\ \tilde{\rho}_h^j(S'_2) &:= (1 - \varepsilon)\rho_h^j(S_2) + \varepsilon\rho_h^j(S_1) \end{aligned}$$

From equation (3) we observe that the choice above leaves the information design part of the revenue unchanged. Thus, the change in revenue from the perturbation is positive whenever

$$\left(\rho_h^t(S_1) + u\rho_h^b(S_1) - \rho_h^t(S_2) - u\rho_h^b(S_2) \right) \left(\int_{S'_1} \left(\frac{1 - F(\mu)}{f(\mu)} \right) dF(\mu) - \int_{S'_2} \left(\frac{1 - F(\mu)}{f(\mu)} \right) dF(\mu) \right) \geq 0$$

As $\rho_h^t(S_2) + u\rho_h^b(S_2) < \rho_h^t(S_1) + u\rho_h^b(S_1)$, the change in revenue due to the perturbation is positive iff

$$\int_{S'_1} \left(\frac{1 - F(\mu)}{f(\mu)} \right) dF(\mu) - \int_{S'_2} \left(\frac{1 - F(\mu)}{f(\mu)} \right) dF(\mu) \geq 0$$

Dividing and multiplying the integrands by μ yields

$$\int_{S'_1} \frac{1 - F(\mu)}{\mu f(\mu)} \mu dF(\mu) - \int_{S'_2} \frac{1 - F(\mu)}{\mu f(\mu)} \mu dF(\mu)$$

By assumption 1 and (MC2) the above is bounded below by

$$\left[\min_{\mu \in S'_1} \left\{ \frac{1 - F(\mu)}{\mu f(\mu)} \right\} - \max_{\mu \in S'_2} \left\{ \frac{1 - F(\mu)}{\mu f(\mu)} \right\} \right] \times \int_{S'_1} \mu dF(\mu) \geq 0$$

This establishes that ρ can be perturbed into a test $\tilde{\rho}$ which preserves obedience and has a non-decreasing slope K . Importantly, the last inequality holds strictly for all smooth cdfs F with support $[0, 1]$ as the function $\frac{1 - F(\mu)}{f(\mu)}$ is non-increasing and $\frac{1}{\mu}$ is strictly decreasing.

A.5 Increased Affinity to Less Selective Receiver

Theorem 1: If assumption 1 holds then there exists a constant $\zeta > 0$ such that if $\zeta > s_b - us_t$ then the optimal mechanism recommends a_t with zero probability ($\rho^t = 0$).

Moreover, under assumption 1, if $s_t > s_b$ then the optimal incentive compatible, obedient and individually rational direct mechanism recommends a_b with positive probability ($\rho_h^b \neq 0$).

Proof. Recall the Lagrangian for the relaxed problem (R) is given by $\mathcal{L}(\rho, \lambda) = \mathbb{E}[\langle l(\mu, \lambda), \rho(\mu) \rangle]$ (see section A.4) where

$$l(\mu, \lambda) = \left(\mu - \frac{1 - F(\mu)}{f(\mu)}, u \left(\mu - \frac{1 - F(\mu)}{f(\mu)} \right), 1 - \mu + \frac{1 - F(\mu)}{f(\mu)}, u \left(1 - \mu + \frac{1 - F(\mu)}{f(\mu)} \right) \right) \\ + (\mu\lambda_t(1 - s_t), \mu\lambda_b(1 - s_b), -(1 - \mu)\lambda_t s_t, -(1 - \mu)\lambda_b s_b)$$

For any optimal mechanism ρ^* and corresponding multiplier λ^* we describe the optimal test choice by it's epigraph.²³

$$\text{epi} \langle l(\cdot, \lambda^*), \rho^*(\cdot) \rangle = \bigcap_{x \in \text{ex}(\mathcal{C})} \text{epi} \langle l(\cdot, \lambda^*), x \rangle$$

The equality follows from the saddle point property. The following conditions are sufficient for a test ρ^* to only assign the sender to the bottom receiver

$$\text{epi} \langle l(\cdot, \lambda^*), (0, 1, 0, 0) \rangle \subseteq \text{epi} \langle l(\cdot, \lambda^*), (1, 0, 0, 0) \rangle$$

$$\text{epi} \langle l(\cdot, \lambda^*), (0, 0, 0, 1) \rangle \subseteq \text{epi} \langle l(\cdot, \lambda^*), (0, 0, u, 0) \rangle$$

$$\text{epi} \langle l(\cdot, \lambda^*), (0, 1, 0, 0) \rangle \subseteq \text{epi} \langle l(\cdot, \lambda^*), (u, 0, 0, 0) \rangle$$

$$\text{epi} \langle l(\cdot, \lambda^*), (0, 1, 0, 1) \rangle \subseteq \text{epi} \langle l(\cdot, \lambda^*), (1, 0, 1, 0) \rangle$$

The above holds simultaneously if the following equations hold simultaneously

$$\lambda_b^*(1 - s_b) - \lambda_t^*(1 - s_t) \geq 1 - u,$$

$$u\lambda_t^* s_t - \lambda_b^* s_b \geq 0,$$

$$\lambda_b^*(1 - s_b) - u\lambda_t^*(1 - s_t) \geq 0,$$

$$\lambda_b^*(\mu - s_b) - \lambda_t^*(\mu - s_t) \geq 1 - u$$

²³The epigraph a function $f : X \rightarrow [-\infty, \infty]$ valued in the extended real numbers is the set $\text{epi } f = \{(x, r) \in X \times \mathbb{R} : r \geq f(x)\}$

Let $s_t > s_b$, and set $\lambda_t^* = \lambda_b^* \frac{s_b}{us_t} + \varepsilon$ for small $\varepsilon > 0$.²⁴ By the choice of λ_t^* and $s_t > s_b$, the second and the third equations above are satisfied. Moreover, the first equation and the fourth hold if the following holds for all μ

$$\lambda_b^* \left(\mu - s_b - (\mu - s_t) \frac{s_b}{us_t} \right) - \varepsilon(\mu - s_t) \geq 1 - u$$

Simplifying gives us the following sufficient condition

$$\lambda_b^* \frac{\mu(us_t - s_b) + (1 - u)s_t s_b}{us_t} - \varepsilon(1 - s_t) \geq 1 - u$$

As ρ^* is a bottom receiver-only allocation, it comprises of tests of the form $(0, 0, 0, 0)$, $(0, 1, 0, 1)$, or $(0, 1, 0, 0)$. By monotonicity of K , the allocation $(0, 1, 0, 0)$ is for higher types than test allocation of $(0, 0, 0, 0)$ or $(0, 1, 0, 1)$. Note that $\langle l(\cdot, \lambda), (0, 0, 0, 0) \rangle = 0$ and $\langle l(\mu, \lambda), (0, 1, 0, 1) \rangle = u + \lambda_b(\mu - s_b)$. In particular, $\langle l(\mu, \lambda), (0, 1, 0, 1) \rangle$ crosses $\langle l(\cdot, \lambda), (0, 0, 0, 0) \rangle$ at most once from below. We get that the optimal test is of the following form

$$(\rho_h^{t,*}(\mu), \rho_h^{b,*}(\mu), \rho_l^{t,*}(\mu), \rho_l^{b,*}(\mu)) = \begin{cases} (0, 0, 0, 0) & \mu \in [0, \mu_0) \\ (0, 1, 0, 1) & \mu \in [\mu_0, \mu_1) \\ (0, 1, 0, 0) & \mu \in [\mu_1, 1] \end{cases} \quad (\rho_{\text{BO}})$$

As $\mathbb{E}[\mu] < q \leq s_b < 1$, we have $0 < \mu_0 < 1$ and $\mu_0 < \mu_1 \leq 1$. By first order necessary conditions for μ_0 , we get

$$\lambda_b^* = \frac{u}{s_b - \mu_0}$$

Positivity of the multipliers implies $\mu_0 < s_b$. Plugging this into the required inequality, we get

$$\frac{\mu(us_t - s_b) + (1 - u)s_t s_b}{s_t(s_b - \mu_0)} - \varepsilon(1 - s_t) \geq 1 - u$$

As $\varepsilon > 0$ can be arbitrarily small, it suffices to consider

$$\frac{\mu(us_t - s_b) + (1 - u)s_t s_b}{s_t(s_b - \mu_0)} > 1 - u$$

Dividing and multiplying by $s_t(s_b - \mu_0)$ and then simplifying the above inequality gives us the following sufficient condition

$$s_t \mu_0 (1 - u) > s_b - us_t$$

Note that the above holds whenever $s_b < us_t$.

²⁴If $s_t \leq s_b$ then the optimal allocation only allocates to the top receiver with positive probability. In particular, this implies that $\zeta < (1 - u)q$.

We have shown that when the above condition holds, then there exists $\lambda^* \geq \mathbb{R}_+^2$ such that the bottom receiver-only allocation (ρ_{BO}) optimizes the Lagrangian pointwise. In particular, (ρ_{BO}) is a solution to (R). The first statement of Theorem 1 follows from setting $\zeta = q \frac{\mu_0^*(1-u)}{2}$. Here μ_0^* is the smallest value for μ_0 across all s_b . This is positive as $s_b \geq q > \mathbb{E}[\mu]$. I present more details about this lower bound in section A.6. The important detail being $\mu_0^* > 0$ and hence $\zeta > 0$.

To prove the second statement, assume for contradiction that for some $s_t > s_b$ the optimal test allocation is top receiver-only. By reasoning similarly to the proof of the first statement, the optimal top receiver-only test is given as following

$$(\rho_h^{t,*}(\mu), \rho_h^{b,*}(\mu), \rho_l^{t,*}(\mu), \rho_l^{b,*}(\mu)) = \begin{cases} (0, 0, 0, 0) & \mu \in [0, \mu_0) \\ (1, 0, 1, 0) & \mu \in [\mu_0, \mu_1) \\ (1, 0, 0, 0) & \mu \in [\mu_1, 1] \end{cases} \quad (\rho_{\text{TO}})$$

Where (ρ_{TO}) pointwise maximizes the Lagrangian $\mathcal{L}(\rho, \lambda) = \mathbb{E}[\langle l(\mu, \lambda), \rho(\mu) \rangle]$ for some choice of multiplier $\lambda^* = (\lambda_t^*, \lambda_b^*)$ where

$$\lambda_t^* = \frac{1}{s_t - \mu_0}$$

and $s_t > \mu_0 > 0$. If $\lambda_b^* = 0$, the unconditional-bottom score equals $u > 0$ on the interval below μ_0 where the proposed top-only allocation rejects. This contradicts pointwise maximization. Hence $\lambda_b^* > 0$.

To establish a contradiction, consider the following two cases:

Case I: If $s_b - \frac{u}{\lambda_b^*} < \mu_0$ then the line $\langle l(\cdot, \lambda^*), (0, 1, 0, 1) \rangle$ crosses $\langle l(\cdot, \lambda^*), (0, 0, 0, 0) \rangle$ before $\langle l(\cdot, \lambda^*), (1, 0, 1, 0) \rangle$ (i.e. before μ_0). Thus (ρ_{TO}) does not pointwise maximize $\mathcal{L}(\cdot, \lambda^*)$.

Case II: If $s_b - \frac{u}{\lambda_b^*} \geq \mu_0$ then the line $\langle l(\cdot, \lambda^*), (0, 1, u, 0) \rangle$ crosses $\langle l(\cdot, \lambda^*), (0, 0, 0, 0) \rangle$ before $\langle l(\cdot, \lambda^*), (1, 0, 1, 0) \rangle$ (i.e. before μ_0). To see this, let μ' be such that $\langle l(\mu', \lambda^*), (0, 1, u, 0) \rangle = 0$. In particular

$$u + \mu'(1 - s_b)\lambda_b^* - u s_t(1 - \mu')\lambda_t^* = 0$$

Rearranging gives the following

$$\mu' \times ((1 - s_b)\lambda_b^* + u s_t \lambda_t^*) = u \frac{\mu_0}{s_t - \mu_0}$$

As $s_b - \frac{u}{\lambda_b^*} \geq \mu_0$ and as $\lambda_t^* = \frac{1}{s_t - \mu_0}$ we get the following

$$\mu' \times \left(\frac{1 - s_b}{s_b - \mu_0} + \frac{s_t}{s_t - \mu_0} \right) \leq \frac{\mu_0}{s_t - \mu_0}$$

As $s_t > s_b$ we get that $\frac{1 - s_b}{s_b - \mu_0} > \frac{1 - s_t}{s_t - \mu_0}$, plugging this in the left-hand side above shows that

$$\mu' < \mu_0$$

Thus (ρ_{TO}) does not pointwise maximize $\mathcal{L}(\cdot, \lambda^*)$ for any value of $\lambda_b^* \in \mathbb{R}_+$. This establishes the second statement of the theorem. $\square$

A.6 Single receiver

Proposition 3: Fix $u = 0$. If $\frac{1-F(\mu)}{f(\mu)}$ is convex and assumption 1 holds, then the top receiver's payoff U_t is increasing in s_t . In particular, the equilibrium standard is such that $s_t = 1$.

Proof. First, note that the receiver never chooses $s_t < \mathbb{E}[\mu] \leq q$. Fix some $s_t \in (q, 1)$. Let $T(s_t) = \mathbb{E}[V(\mu)]$ be the expected gross utility generated by the optimal mechanism corresponding to standard s_t . As the obedience constraint must bind for the optimal mechanism, the receiver's payoff is given by:

$$(s_t v_h + (1 - s_t) v_l) T(s_t)$$

The receiver prefers a standard $s'_t > s_t$ over s_t if and only if

$$\begin{aligned} & (s'_t v_h + (1 - s'_t) v_l) T(s'_t) - (s_t v_h + (1 - s_t) v_l) T(s_t) > 0 \\ \iff & s'_t T(s'_t) - s_t T(s_t) > q (T(s'_t) - T(s_t)) \end{aligned} \quad (5)$$

As argued in section A.5, under assumption 1, the optimal mechanism can be found among the ones with the following structure:

$$(\rho_h^t(\mu), \rho_l^t(\mu)) = \begin{cases} (0, 0) & \mu \in [0, \mu_0) \\ (1, 1) & \mu \in [\mu_0, \mu_1) \\ (1, 0) & \mu \in [\mu_1, 1] \end{cases}$$

Where $\int_{\mu_0}^1 \mu dF(\mu) = \frac{s_t}{1-s_t} \int_{\mu_0}^{\mu_1} (1 - \mu) dF(\mu)$.

For a given s_t , the optimal mechanism can be described by the tuple $(\mu_0(s_t), \mu_1(s_t))$. Consider the Lagrangian, $\mathcal{L}(\mu_0, \mu_1, \lambda; s_t)$, corresponding to the designer's constrained optimization problem, given by

$$\int_{\mu_0}^{\mu_1} (1 + \lambda(\mu - s_t)) dF(\mu) + \int_{\mu_1}^1 \left(\mu(1 + \lambda(1 - s_t)) - \frac{1 - F(\mu)}{f(\mu)} \right) dF(\mu)$$

Let $\mu_0(s_t)$, $\mu_1(s_t)$, $\lambda(s_t)$ be the optimal solutions. Note that $\lambda(s_t) \geq 0$ and by assumption 1 the inverse hazard rate $\frac{1-F(\mu)}{f(\mu)}$ is decreasing.

First, we will show that $\mu_0(s_t)$ is increasing. Using the first order necessary conditions with respect to μ_0 and μ_1 we get the following

$$\begin{aligned} \lambda(s_t) &= \frac{1}{s_t - \mu_0(s_t)} \\ \lambda(s_t) &= \frac{1}{s_t} \left[1 + \frac{1 - F(\mu_1(s_t))}{(1 - \mu_1(s_t)) f(\mu_1(s_t))} \right] \end{aligned}$$

First, I will handle the case when there is a corner solution with $\mu_1(s_t) = 1$. Only the first order necessary condition for μ_0 holds. By positivity of the multiplier, we get $\mu_0(s_t) < s_t$. By the saddle

point property of the Lagrangian, we must have $1 + \lambda(\mu - s_t) \geq \mu(1 + \lambda(1 - s_t)) - \frac{1-F(\mu)}{f(\mu)}$ for all $\mu \in [0, 1]$. Importantly, $1 + \lambda(\mu - s_t)$ becomes positive before $\mu(1 + \lambda(1 - s_t)) - \frac{1-F(\mu)}{f(\mu)}$ turns positive. As $1 + \lambda(\mu - s_t)$ is positive for $\mu > \mu_0$ and as $\mu(1 + \lambda(1 - s_t)) - \frac{1-F(\mu)}{f(\mu)} \geq \mu - \frac{1-F(\mu)}{f(\mu)}$, we get that μ_0 is smaller than μ' where $\mu' - \frac{1-F(\mu')}{f(\mu')} = 0$. I will show later that this implies $\mu_0 < \mathbb{E}_F[\mu]$.

If the top receiver marginally increases its standard from s_t to $s'_t > s_t$ and $\mu_1(s'_t) = 1$. For obedience to hold $\mu_0(s'_t) > \mu_0(s_t)$. The new mechanism removes types with the lowest expected quality, which is below $\mathbb{E}_F[\mu] < q$, thus the top receiver's payoff increases.

Now, let's consider the case with an interior solution. Define the following function:

$$Q_1(\mu_0, \mu_1; s_t) := \frac{s_t - \mu_0}{s_t} \left[1 + \frac{1 - F(\mu_1)}{(1 - \mu_1)f(\mu_1)} \right] - 1$$

Now, consider the obedience constraint

$$(1 - s_t) \int_{\mu_0(s_t)}^1 \mu dF(\mu) - s_t \int_{\mu_0(s_t)}^{\mu_1(s_t)} (1 - \mu) dF(\mu) = 0$$

Define the following function of (μ_0, μ_1, s_t)

$$Q_2(\mu_0, \mu_1; s_t) := (1 - s_t) \int_{\mu_0}^1 \mu dF(\mu) - s_t \int_{\mu_0}^{\mu_1} (1 - \mu) dF(\mu)$$

Let $Q(\mu_0, \mu_1; s_t) := (Q_1, Q_2)$. By the first order condition and binding obedience constraint, we get

$$Q(\mu_0(s_t), \mu_1(s_t); s_t) = (0, 0)$$

The Jacobian of Q is given by

$$J(\mu_0, \mu_1; s_t) := \begin{bmatrix} \frac{\partial}{\partial s_t} Q_1 & \frac{\partial}{\partial \mu_0} Q_1 & \frac{\partial}{\partial \mu_1} Q_1 \\ \frac{\partial}{\partial s_t} Q_2 & \frac{\partial}{\partial \mu_0} Q_2 & \frac{\partial}{\partial \mu_1} Q_2 \end{bmatrix} = \begin{bmatrix} J_s & J_\mu \end{bmatrix}$$

Where

$$J_s = \begin{bmatrix} \frac{\mu_0}{s_t^2} \left[1 + \frac{1-F(\mu_1)}{(1-\mu_1)f(\mu_1)} \right] \\ - \int_{\mu_0}^1 \mu dF(\mu) - \int_{\mu_0}^{\mu_1} (1 - \mu) dF(\mu) \end{bmatrix}$$

and

$$J_\mu = \begin{bmatrix} -\frac{1}{s_t} \left[1 + \frac{1-F(\mu_1)}{(1-\mu_1)f(\mu_1)} \right] & \frac{s_t - \mu_0}{s_t} \frac{\partial}{\partial \mu_1} \left(\frac{1-F(\mu_1)}{(1-\mu_1)f(\mu_1)} \right) \\ (s_t - \mu_0)f(\mu_0) & -s_t(1 - \mu_1)f(\mu_1) \end{bmatrix}$$

From the above, we get

$$\det J_\mu = (1 - \mu_1)f(\mu_1) \left[1 + \frac{1 - F(\mu_1)}{(1 - \mu_1)f(\mu_1)} \right] - \frac{(s_t - \mu_0)^2}{s_t} f(\mu_0) \frac{\partial}{\partial \mu_1} \left(\frac{1 - F(\mu_1)}{(1 - \mu_1)f(\mu_1)} \right)$$

Assumption 1 implies $\frac{\partial}{\partial \mu} \left(\frac{1-F(\mu)}{(1-\mu)f(\mu)} \right) \leq 0$, thus

$$\det J_\mu |_{(\mu_0(s_t), \mu_1(s_t), s_t)} > 0$$

By Implicit Function Theorem $(\mu_0(s_t), \mu_1(s_t))$ can be described by an implicit function g . In particular $g(s_t) = (\mu_0(s_t), \mu_1(s_t))$ and

$$\frac{\partial}{\partial s_t} g = -J_\mu^{-1} \times J_s$$

Where

$$J_\mu^{-1} = \frac{1}{\det J_\mu} \begin{bmatrix} -s_t(1-\mu_1)f(\mu_1) & -\frac{s_t-\mu_0}{s_t} \frac{\partial}{\partial \mu_1} \left(\frac{1-F(\mu_1)}{(1-\mu_1)f(\mu_1)} \right) \\ -(s_t-\mu_0)f(\mu_0) & -\frac{1}{s_t} \left[1 + \frac{1-F(\mu_1)}{(1-\mu_1)f(\mu_1)} \right] \end{bmatrix}$$

We can conclude that $\frac{\partial}{\partial s_t} \mu_0(s_t) \geq 0$ as

$$\begin{aligned} \frac{\mu_0(s_t)}{s_t} (1-\mu_1(s_t)) f(\mu_1(s_t)) \left[1 + \frac{1-F(\mu_1(s_t))}{(1-\mu_1(s_t))f(\mu_1(s_t))} \right] \\ - T(s_t) \frac{s_t-\mu_0(s_t)}{s_t} \frac{\partial}{\partial \mu_1} \left(\frac{1-F(\mu_1(s_t))}{(1-\mu_1(s_t))f(\mu_1(s_t))} \right) > 0 \end{aligned}$$

This establishes that $\mu_0(s_t)$ is increasing.

Next, we will show that $\mu_1(s_t)$ is decreasing. After some algebra and using the fact that the obedience constraint binds, we get $\frac{\partial}{\partial s_t} \mu_1(s_t)$ equals

$$\frac{1}{\det J_\mu} \left[\frac{1}{s_t^2} \left(1 + \frac{1-F(\mu_1(s_t))}{(1-\mu_1(s_t))f(\mu_1(s_t))} \right) \left(\mu_0(s_t)(s_t-\mu_0(s_t))f(\mu_0(s_t)) - \int_{\mu_0(s_t)}^1 \mu dF(\mu) \right) \right]$$

The sign of $\frac{\partial}{\partial s_t} \mu_1(s_t)$ is determined by

$$\begin{aligned} \mu_0(s_t)(s_t-\mu_0(s_t))f(\mu_0(s_t)) - \int_{\mu_0(s_t)}^1 \mu dF(\mu) \\ < \mu_0(s_t) \times ((1-\mu_0(s_t))f(\mu_0(s_t)) - 1 + F(\mu_0(s_t))) \end{aligned}$$

Assumption 1 implies the bounded hazard rate property. Thus, we get

$$(1-\mu_0(s_t))f(\mu_0(s_t)) - 1 + F(\mu_0(s_t)) \leq 0$$

$$\implies \frac{\partial}{\partial s_t} \mu_1(s_t) \leq 0$$

This establishes that $\mu_1(s_t)$ is decreasing.

By the first order condition for μ_1 , we get the following

$$\mu_1(s_t) (1 + (1-s_t) \lambda(s_t)) - \frac{1-F(\mu_1(s_t))}{f(\mu_1(s_t))} = 1 + (\mu_1(s_t) - s_t) \lambda(s_t) \quad (6)$$

By the binding obedience constraint, we get that $\mu_0(1) = \mu_1(1)$. Using this and evaluating equation (6) for $s_t = 1$ we get

$$\mu_1(1) - \frac{1 - F(\mu_1(1))}{f(\mu_1(1))} = 0 \quad (7)$$

By the inverse-hazard convexity hypothesis of the proposition, $\frac{1-F(\mu)}{f(\mu)}$ is convex and we can apply Jensen inequality to claim the following

$$\begin{aligned} \mathbb{E}[\mu] &= \mathbb{E} \left[\frac{1 - F(\mu)}{f(\mu)} \right] \geq \frac{1 - F(\mathbb{E}[\mu])}{f(\mathbb{E}[\mu])} \\ \implies \mathbb{E}[\mu] - \frac{1 - F(\mathbb{E}[\mu])}{f(\mathbb{E}[\mu])} &\geq 0 \end{aligned}$$

Moreover, $\mu - \frac{1-F(\mu)}{f(\mu)}$ is increasing by assumption 1. Equation (7) then implies that $\mu_0(1) = \mu_1(1) \leq \mathbb{E}[\mu]$. Monotonicity of $\mu_0(s_t)$ then establishes $\mu_0(s_t) < \mathbb{E}[\mu]$ for all s_t .

The designer responds to a marginal increase in standards by increasing μ_0 and lowering μ_1 . As $\mu_0 < q$, the average quality of senders excluded by the designer is low enough. In particular, equation (5) holds for $s'_t = 1$ and for all $s_t \in [q, 1)$.

To complete the argument, we need to show that $\mu_0(s_t) > 0$ for all $s_t > \mathbb{E}[\mu]$. This rules out the corner solution with $\mu_0(s_t) = 0$. As $s_t > \mathbb{E}[\mu]$, by obedience constraint we get

$$\mu_0(s_t) = 0 \implies \mu_1(s_t) < 1$$

Using the first-order necessary conditions, we get

$$\frac{1}{s_t} \geq \frac{1}{s_t} \left[1 + \frac{1 - F(\mu_1(s_t))}{(1 - \mu_1(s_t))f(\mu_1(s_t))} \right]$$

This leads to a contradiction as $1 + \frac{1-F(\mu_1(s_t))}{(1-\mu_1(s_t))f(\mu_1(s_t))} > 1$. This also establishes the required lower bound in section A.5 ($\mu_0^* > 0$).

□

A.7 Equilibrium Well-posedness

We first show that in any pure strategy equilibrium the top receiver's standard $s_t \geq s_b$. Assume for contradiction that $s_t < s_b$, then the designer chooses to allocate to only the top receiver. If $s_t < s_b$, any test allocation ρ is revenue equivalent to a perturbed test allocation $\hat{\rho}$. Where $\hat{\rho}$ is defined such that for all μ and θ , the allocation $\hat{\rho}$ recommends the sender to the top receiver with probability $\rho_\theta^t(\mu) + u\rho_\theta^b(\mu)$ and rejects with probability $1 - \rho_\theta^t(\mu) - u\rho_\theta^b(\mu)$. This introduces slack to the top receivers' obedience constraint, as $s_b > s_t$, thus ρ is not optimal. By section A.6, the top receiver has a profitable deviation of setting $s_t = s_b$. This shows there is no equilibrium (in pure strategy) for which $s_t < s_b$.

Now we will prove the continuity of the receiver payoffs, which will allow us to establish the existence of a mixed strategy equilibrium.

Lemma 2: Let F satisfy assumption 1. For each $(s_t, s_b) \in [q, 1]^2$, the receiver payoffs U_t and U_b are constant over the set of solutions to the primal problem (P). Moreover, the payoffs U_t and U_b vary continuously with (s_t, s_b) .

Proof. I present the proof in two parts. First, I will establish the conditions for Berge's Maximum Theorem (see chapter 17 in Aliprantis and Border (2006) for reference). This will establish upper hemicontinuity of the relaxed optimizer correspondence. Proposition 2 identifies its attainable receiver payoffs with those of the primal problem. Second, I will show that for any given $s = (s_t, s_b)$, the receivers' utility U_t and U_b are constant over the set of optimal mechanisms. This, along with upper hemicontinuity of the solution set and continuity of U_t, U_b in the test allocation ρ implies continuity of U_t and U_b in s .

Claim 1: The relaxed value is continuous, and the relaxed optimizer correspondence is nonempty, compact-valued, and upper hemicontinuous.

Recall the weakly compact convex set $\mathcal{F}$ from Appendix A.4. For $j \in \{t, b\}$, write

$$H_j(\rho) = \int_0^1 \mu \rho_h^j(\mu) dF(\mu), \quad L_j(\rho) = \int_0^1 (1 - \mu) \rho_l^j(\mu) dF(\mu),$$

and $G_j(s, \rho) = (1 - s_j)H_j(\rho) - s_j L_j(\rho)$. Use the inequality formulation of the relaxed feasible correspondence:

$$\mathcal{O}(s) = \{\rho \in \mathcal{F} : G_t(s, \rho) \geq 0, G_b(s, \rho) \geq 0\}.$$

The zero allocation belongs to $\mathcal{O}(s)$ for every s . Each G_j is jointly continuous in standards and the weak topology on $\mathcal{F}$, because H_j and L_j are bounded linear functionals on $L_2(F)^4$. Thus $\mathcal{O}$ has a closed graph and compact values in the common compact range $\mathcal{F}$, which implies upper hemicontinuity.

To establish lower hemicontinuity, fix $\rho \in \mathcal{O}(s)$ and a sequence $s^n \rightarrow s$. Let

$$z = (1/2, 1/2, 0, 0) \in \mathcal{F}.$$

For an index with $s_j = 1$, feasibility implies $L_j(\rho) = 0$ and hence

$$G_j(s^n, \rho) = (1 - s_j^n)H_j(\rho) \geq 0.$$

The allocation z also satisfies this inequality. For the remaining indices, $G_j(s^n, z) = (1 - s_j^n)\mathbb{E}_F[\mu]/2$ is bounded below by a common $c > 0$ for all sufficiently large n . Define

$$d_n = \max_{j: s_j < 1} [-G_j(s^n, \rho)]_+, \quad \varepsilon_n = \frac{d_n}{c + d_n}, \quad \rho^n = (1 - \varepsilon_n)\rho + \varepsilon_n z,$$

where $[x]_+ = \max\{x, 0\}$. Then $d_n \rightarrow 0$ and, for each index with $s_j < 1$,

$$G_j(s^n, \rho^n) \geq -(1 - \varepsilon_n)d_n + \varepsilon_n c = 0.$$

The indices with $s_j = 1$ are already satisfied by both components of the mixture. Consequently $\rho^n \in \mathcal{O}(s^n)$ and $\rho^n \rightarrow \rho$ in norm. If both standards equal one, take $\rho^n = \rho$ directly. This proves lower hemicontinuity, including all boundary cases. The sequential argument suffices because the weak topology is metrizable on bounded subsets of the separable Hilbert space $L_2(F)^4$.

Using the weak continuity of the relaxed objective on $\mathcal{F}$ stated in Appendix A.4, the objective is continuous on this common compact range. Berge's Maximum Theorem therefore gives the stated continuity of the value and upper hemicontinuity of the optimizer correspondence $X^*(s)$. By Proposition 2, every relaxed optimizer has a primal representative with the same receiver moments, and every primal optimizer is relaxed optimal. The two programs therefore induce the same sets of receiver payoffs. The remaining step is to show that each such set is a singleton.

Claim 2: For any given $s = (s_t, s_b)$, the receivers' utility U_t and U_b are constant over the set of optimal mechanisms.

Fix some (s_t, s_b) . Represent the set of solutions to (R) by $X^*(s)$. Recall for an optimal test allocation ρ the obedience constraint binds. Thus for any $\rho \in X^*(s)$ we get

$$\begin{aligned} U_t(\rho) &= \left(v_h + v_l \frac{1 - s_t}{s_t} \right) \mathbb{E} [\mu \rho_h^t(\mu)] \\ U_b(\rho) &= \left(v_h + v_l \frac{1 - s_b}{s_b} \right) \mathbb{E} [\mu \rho_h^b(\mu)] \end{aligned}$$

In particular, U_t and U_b are constant over $X^*(s)$ if the following holds

$$\rho, \rho' \in X^*(s) \implies \rho_h = \rho'_h \quad (8)$$

I will show that the above statement (8) holds by means of contradiction. Assume that there exist $\rho, \rho' \in X^*(s)$ such that $\rho_h \neq \rho'_h$. In particular, there exists a set $S \subset [0, 1]$ of types with positive measure such that $\rho_h(\mu) \neq \rho'_h(\mu)$ for all $\mu \in S$. Note that $X^*(s)$ is convex so $\alpha\rho + (1 - \alpha)\rho' \in X^*(s)$ for all $\alpha \in [0, 1]$. We will proceed in two cases:

Case I: There exists a positive probability subset $S' \subset S$ such that for all $\mu \in S'$ either $K(\mu; \rho) > 0$ or $K(\mu; \rho') > 0$. In this case, we consider some $\alpha \in (0, 1)$ and define $\hat{\rho} := \alpha\rho + (1 - \alpha)\rho'$. On the set S' , we have $0 < \hat{\rho}_h^t, \hat{\rho}_h^b < 1$ and $K(\cdot; \hat{\rho}) > 0$. The previous statement follows from the characterization of extreme points. We can then choose two positive probability (ordered) subsets $S'_1 < S'_2 \subset S'$. Finally, we perturb $\hat{\rho}$ on S'_1 by slightly increasing $\hat{\rho}_h^b$ and slightly decreasing $\hat{\rho}_h^t$. To maintain obedience, we also perturb $\hat{\rho}$ on S'_2 by slightly increasing $\hat{\rho}_h^t$ and slightly decreasing $\hat{\rho}_h^b$. A calculation similar to proof of Proposition 3, in section A.4, shows that constructing such a perturbation without violating obedience is feasible and yields strictly greater revenue, a contradiction.

Case II: For all $\mu \in S$ the slope $K(\mu; \rho) = K(\mu; \rho') = 0$. Define $\hat{\rho} := \alpha\rho + (1 - \alpha)\rho'$ and $\bar{\mu} := \inf\{\mu \mid K(\mu, \hat{\rho}) > 0\}$. We can partition $[0, \bar{\mu}]$ into intervals $[\mu_i, \mu_{i+1})$ for $0 = \mu_0 \leq \mu_1 \leq \mu_2 \leq \mu_3 \leq \mu_4 \leq \bar{\mu}$. Where

$$\begin{aligned} \int_{\mu_4}^{\bar{\mu}} \mu dF(\mu) &= \alpha \int_0^{\bar{\mu}} \mu \cdot 1_{\{\rho=(1,0,1,0)\}} dF(\mu) + (1 - \alpha) \int_0^{\bar{\mu}} \mu \cdot 1_{\{\rho'=(1,0,1,0)\}} dF(\mu) \\ \int_{\mu_3}^{\mu_4} \mu dF(\mu) &= \alpha \int_0^{\bar{\mu}} \mu \cdot 1_{\{\rho=(0,1,u,0)\}} dF(\mu) + (1 - \alpha) \int_0^{\bar{\mu}} \mu \cdot 1_{\{\rho'=(0,1,u,0)\}} dF(\mu) \\ \int_{\mu_2}^{\mu_3} \mu dF(\mu) &= \alpha \int_0^{\bar{\mu}} \mu \cdot 1_{\{\rho=(u,0,0,1)\}} dF(\mu) + (1 - \alpha) \int_0^{\bar{\mu}} \mu \cdot 1_{\{\rho'=(u,0,0,1)\}} dF(\mu) \\ \int_{\mu_1}^{\mu_2} \mu dF(\mu) &= \alpha \int_0^{\bar{\mu}} \mu \cdot 1_{\{\rho=(0,1,0,1)\}} dF(\mu) + (1 - \alpha) \int_0^{\bar{\mu}} \mu \cdot 1_{\{\rho'=(0,1,0,1)\}} dF(\mu) \end{aligned}$$

The partition constructed above rearranges the mass of high quality ($\theta = h$) senders for types below $\bar{\mu}$. Define $\tilde{\rho}$ which equals $\hat{\rho}$ for types greater than $\bar{\mu}$ and defined as following for types $\mu \leq \bar{\mu}$

$$\tilde{\rho}(\mu) = \begin{cases} (0,0,0,0) & \mu < \mu_1 \\ (0,1,0,1) & \mu \in [\mu_1, \mu_2) \\ (u,0,0,1) & \mu \in [\mu_2, \mu') \\ (u,0,u,0) & \mu \in [\mu', \mu_3) \\ (0,1,u,0) & \mu \in [\mu_3, \mu_4) \\ (1,0,1,0) & \mu \in [\mu_4, \bar{\mu}] \end{cases} \quad \text{or} \quad \begin{cases} (0,0,0,0) & \mu < \mu_1 \\ (0,1,0,1) & \mu \in [\mu_1, \mu') \\ (0,1,u,0) & \mu \in [\mu', \mu_2) \\ (u,0,u,0) & \mu \in [\mu_2, \mu_3) \\ (0,1,u,0) & \mu \in [\mu_3, \mu_4) \\ (1,0,1,0) & \mu \in [\mu_4, \bar{\mu}] \end{cases}$$

Where μ' is chosen such that

$$\begin{aligned} \int_0^{\bar{\mu}} (1 - \mu) \cdot \left(1_{\{\tilde{\rho}=(0,1,0,1)\}} + 1_{\{\tilde{\rho}=(u,0,0,1)\}} \right) dF(\mu) \\ = \alpha \int_0^{\bar{\mu}} (1 - \mu) \cdot 1_{\{\rho=(0,1,0,1)\}} dF(\mu) + (1 - \alpha) \int_0^{\bar{\mu}} (1 - \mu) \cdot 1_{\{\rho'=(0,1,0,1)\}} dF(\mu) \end{aligned}$$

By construction, we get

$$\int_0^1 (1 - \mu) \tilde{\rho}_l^b(\mu) dF(\mu) = \int_0^1 (1 - \mu) \hat{\rho}_l^b(\mu) dF(\mu)$$

and

$$\int_0^1 \mu \tilde{\rho}_h^j(\mu) dF(\mu) = \int_0^1 \mu \hat{\rho}_h^j(\mu) dF(\mu)$$

for $j \in \{t, b\}$.

When $\rho \neq \rho'$, the allocation $\tilde{\rho}$ induces slack in the top receiver's obedience constraint. The test allocation $\tilde{\rho}$ is constructed such that the mass of low quality senders matched with the bottom receiver is preserved, and the mass of low quality senders matched with the top receiver is pushed

forward to higher types. As $(1 - \mu)$ is decreasing, the test allocation $\tilde{\rho}$ leads to a strictly lower mass of low quality senders matched with the top receiver relative to $\hat{\rho}$. We can construct a perturbation $\rho'' := \beta\tilde{\rho} + (1 - \beta)(1, 0, 1, 0)$ for $\beta \in (0, 1)$. We can choose β so that the top receiver's obedience constraint binds, leading to a contradiction as the allocation ρ'' yields a greater revenue than $\hat{\rho}$. $\square$

By the Glicksberg's theorem, see [Fudenberg and Tirole \(1991\)](#), continuity of the first period payoffs and compact choice sets implies the existence of an equilibrium with possibly mixed strategies played by the receivers.

A.8 Proof of Theorem 2

Proof. Receiver participation implies $\bar{U}_t(\sigma), \bar{U}_b(\sigma) \geq 0$: each receiver can ensure a nonnegative payoff by choosing a standard of 1.

Equilibrium standards are at least q . The top receiver never chooses $s_t \leq \mathbb{E}_F[\mu]$. At such a standard, the certifier can sell an uninformative test that assigns every sender to the top receiver, charging the sender's entire gain over not purchasing. Revenue maximization then assigns everyone to that receiver, whose payoff is $(v_h - v_l)(\mathbb{E}_F[\mu] - q) < 0$. Thus $s_t > \mathbb{E}_F[\mu]$ almost surely.

By binding obedience in [Equation 2](#), matches to the top receiver have expected quality s_t , so choosing $s_t < q$ gives it a negative payoff whenever matched. The bottom receiver likewise loses money whenever matched at $\mathbb{E}_F[\mu] \leq s_b < q$, since its matched expected quality equals s_b .

Suppose $s_b < \mathbb{E}_F[\mu]$ but the bottom receiver's matched expected quality exceeds the prior mean. Raising s_b strictly between the prior mean and that quality leaves the original mechanism feasible. It also remains optimal: a small lottery with any better mechanism at the higher standard would preserve the original constraints and increase revenue. Rejection has positive probability by Bayes plausibility, so its strict obedience constraint survives a sufficiently small lottery. The original mechanism would therefore be optimal with matched quality strictly above the new standard, contradicting [Equation 2](#). Hence the bottom receiver's matched expected quality is at most $\mathbb{E}_F[\mu] < q$, giving it a negative payoff.

Consequently, a receiver choosing below q loses money whenever it receives matches. Participation and best-response optimality rule out such matches in equilibrium. If $s_t < q$, either $s_b < s_t$ and the bottom receiver receives matches by [Theorem 1](#), or $s_b \geq s_t$ and only the top receiver receives matches. The latter matching mass is positive because a priced high-quality-only test yields positive certifier revenue. Either case gives matches to a receiver choosing below q , a contradiction. Hence $s_t \geq q$ almost surely. Any $s_b < q$ would now give the bottom receiver matches and a loss, so $s_b \geq q$ almost surely as well. On these standards, the definition of q and [Equation 2](#) give

$$U_j(s, \rho) = (v_h - v_l)(s_j - q)P_j(s, \rho), \quad j \in \{t, b\}.$$

The bottom receiver does not always choose q . Suppose $\sigma_b(\{q\}) = 1$, so its payoff is zero. If the top receiver chooses above q with positive probability, then, for some sufficiently small $\varepsilon > 0$, choosing $s'_b = q + \varepsilon$ gives the bottom receiver matches whenever $s_t > s'_b$ by [Theorem 1](#). This event has positive probability and its payoff is nonnegative elsewhere, yielding a profitable deviation. Hence the top receiver must also choose q with probability one.

At (q, q) , both receivers earn zero. The top receiver can profitably deviate to $s'_t = q + \varepsilon$ for sufficiently small $\varepsilon > 0$. To see this, write $\Pi(s_t, s_b)$ for optimal certifier revenue. Selling a high-quality-only top receiver test $(\rho_h^t, \rho_l^t) = (1, 0)$ at a sufficiently small positive price gives $\Pi(q, q) > 0$. Any mechanism excluding the top receiver at $(q + \varepsilon, q)$ earns at most $u\Pi(q, q)$: assigning its matches to the top receiver and dividing its prices by u gives a feasible mechanism at (q, q) . Continuity of optimal revenue, established in the proof of [Lemma 2](#), and $u < 1$ imply

$$\Pi(q + \varepsilon, q) > u\Pi(q, q)$$

for all sufficiently small $\varepsilon > 0$. Every optimal allocation therefore gives the top receiver matches and a strictly positive payoff. This contradicts equilibrium at (q, q) , ruling out $\sigma_b(\{q\}) = 1$.

The top receiver earns positive payoff and receives matches. Suppose $\bar{U}_t(\sigma) = 0$. The bottom receiver chooses above q with positive probability. For sufficiently small $\varepsilon > 0$, the top receiver can choose $s'_t = q + \varepsilon$ with $\Pr_{\sigma_b}(s_b > s'_t) > 0$. On this event only the top receiver receives matches, and its payoff is positive; elsewhere its payoff is nonnegative. This profitable deviation contradicts $\bar{U}_t(\sigma) = 0$. Hence $\bar{U}_t(\sigma) > 0$ and $\bar{P}_t(\sigma) > 0$.

The bottom receiver also earns positive payoff and receives matches. Since choosing $s_t = q$ yields zero, the top receiver chooses above q with positive probability. For sufficiently small $\varepsilon > 0$, the bottom receiver can choose $s'_b = q + \varepsilon$ with $\Pr_{\sigma_t}(s_t > s'_b) > 0$. By [Theorem 1](#), it receives matches and positive payoff on this event, and nonnegative payoff elsewhere. Hence $\bar{U}_b(\sigma) > 0$ and $\bar{P}_b(\sigma) > 0$. Both matching shares in [Equation 4](#) therefore lie strictly between zero and one, giving the comparison with the benchmark in [Proposition 1](#). $\square$